



Building engineering

Porsche showroom



ENERGY AUDIT REPORT

Date 7th May 2024

PORSCHE



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INTRODUCTION

Analysing the energy dynamics within the building, scrutinizing its ingress and utilization, and identifying avenues for enhancement and energy efficiency measures.



Porsche centre Khobar, Khobar Dammam highway



1759



2010



GOALS

Main Goal

- Optimize energy consumption within the building.

Secondary Goal

- Achieve savings on electricity usage

OBJECTIVES

- Perform a simple **walk-through audit** of electrical appliances' energy usage in the showroom.
- Review and analyse historical energy usage data to establish a baseline for potential savings calculations.
- Identify opportunities to **lower energy usage** throughout the building.
- Explore **financing options** for system improvements, if available.
- Propose actions to **enhance** the building's environmental **performance**.

Purpose and scope of the report

The report endeavors to assess and enhance the existing level of energy consumption within the building. Investigations will be undertaken to identify building issues contributing to excessive energy usage, followed by the development of solutions. These solutions will then be implemented within a simulation program, such as Design Builder, to gauge the extent of improvement in the building's energy efficiency.



GENERAL INFORMATION

Building data

Building type	Car showroom
Location	Porsche centre Khobar, Khobar Dammam highway

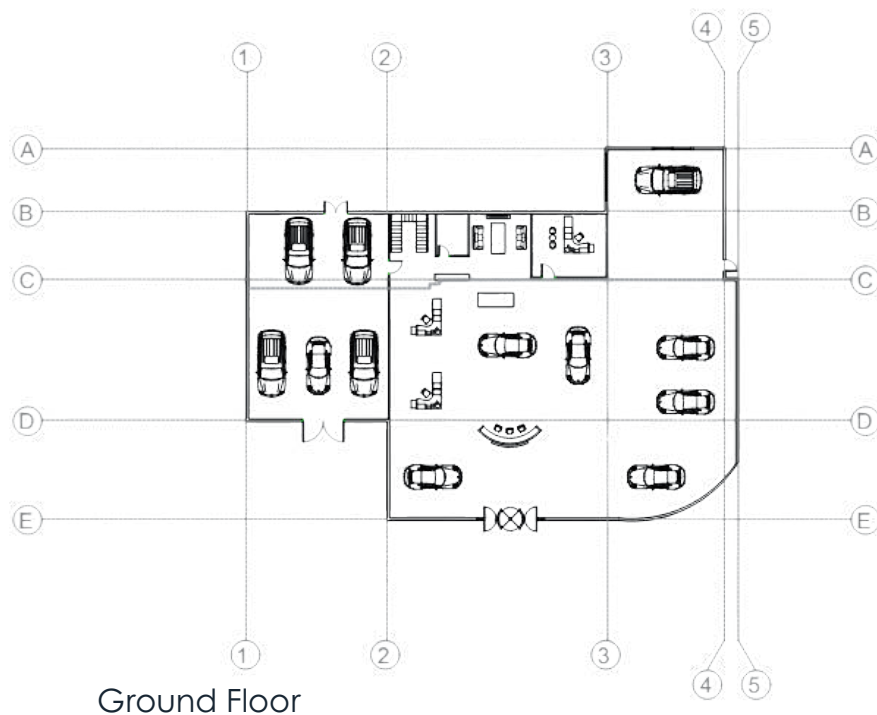
Description

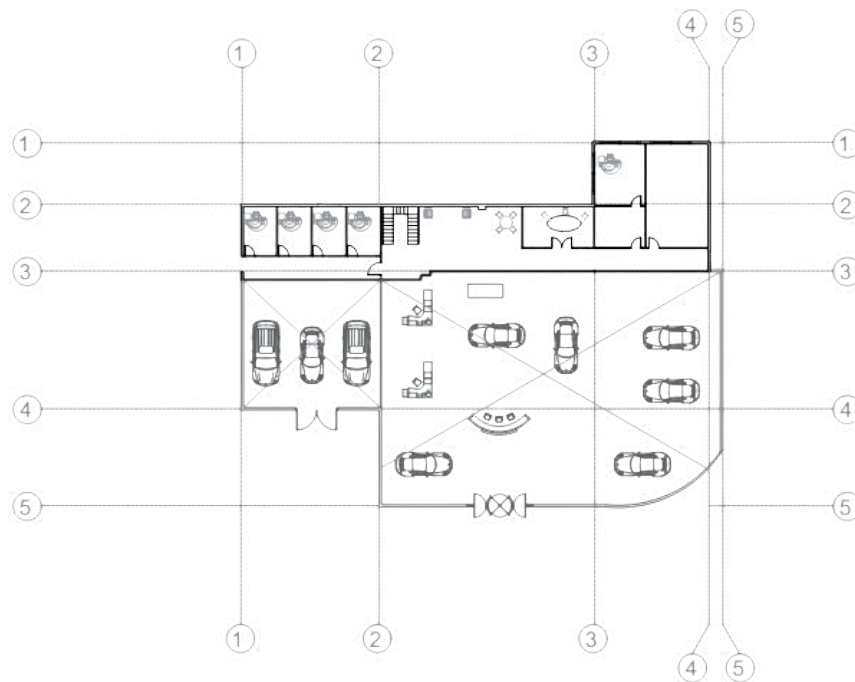
Ground floor	1234
First floor	525
Total area	1759
Building age	14 years

Services

Glazing type	High performance double glazed
Electrical system	3 phase
Lighting system	LED, spoghtlights, and strip lights

PLANS & ELEVATIONS



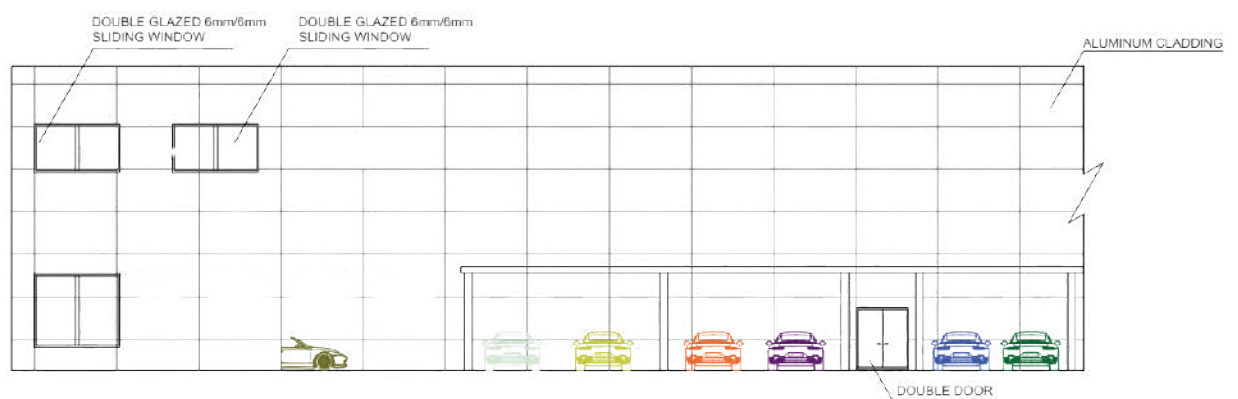


Mezzanine Floor

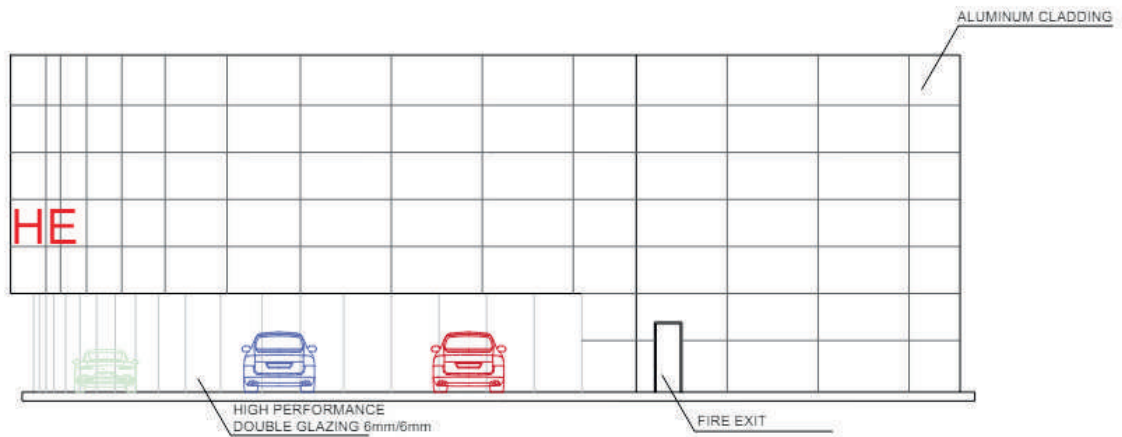
ELEVATIONS



North-east elevation



South-west elevation



North-west elevation

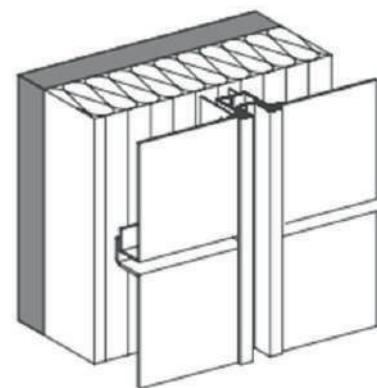
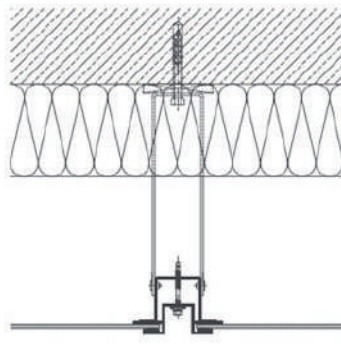
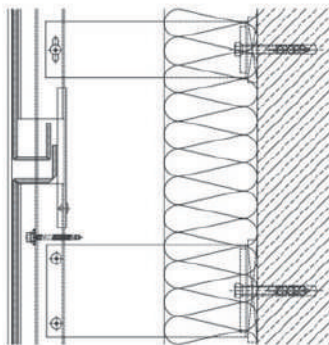
ENERGY FORM USED

- Electricity
- Lighting
- Equipment
- Heating and Ventilation Air Condition - HVAC-

BUILDING CONSTRUCTION MATERIALS

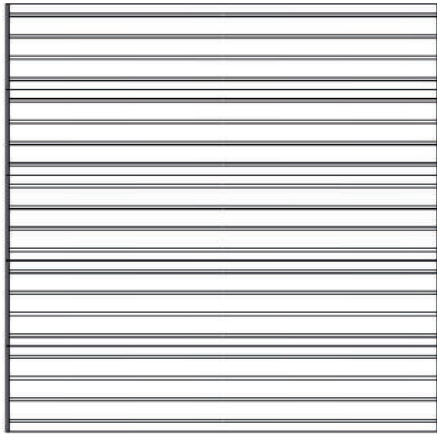
- Alucobond by Pilakoutas

ALUCOBOND®

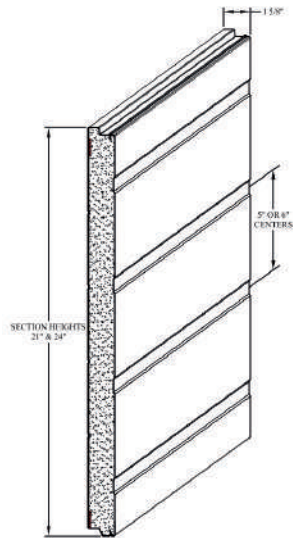




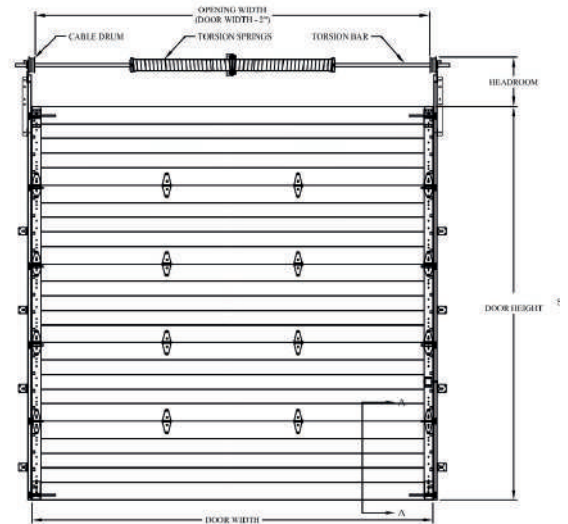
- Garage door



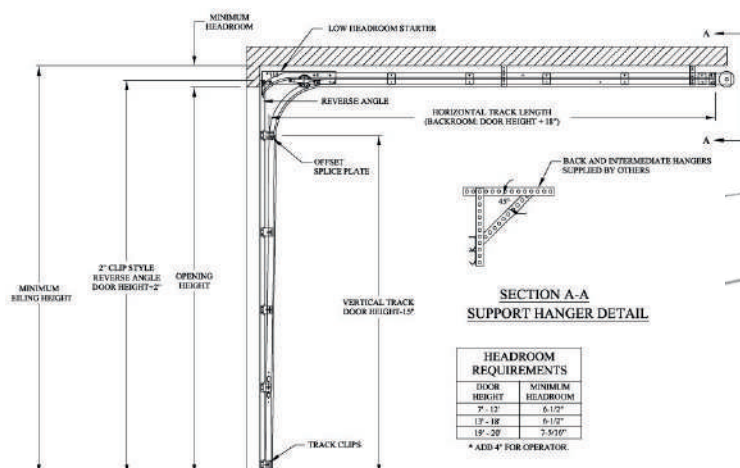
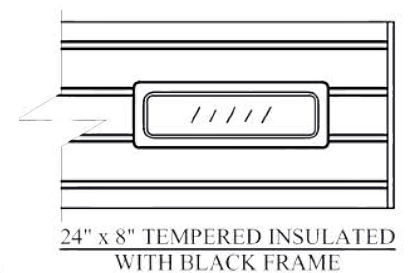
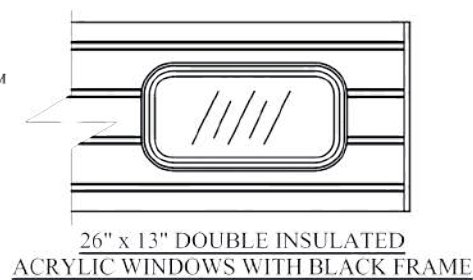
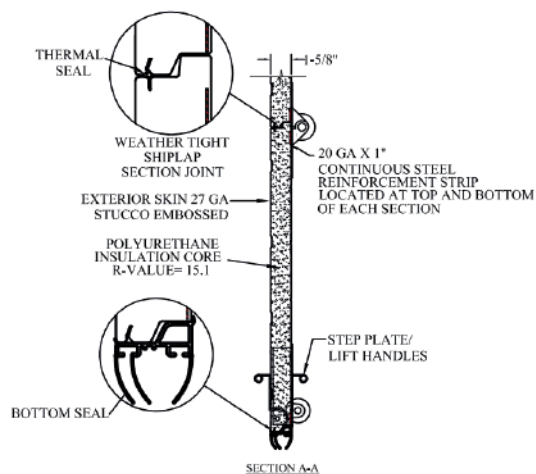
Exterior elevation



Interior elevation

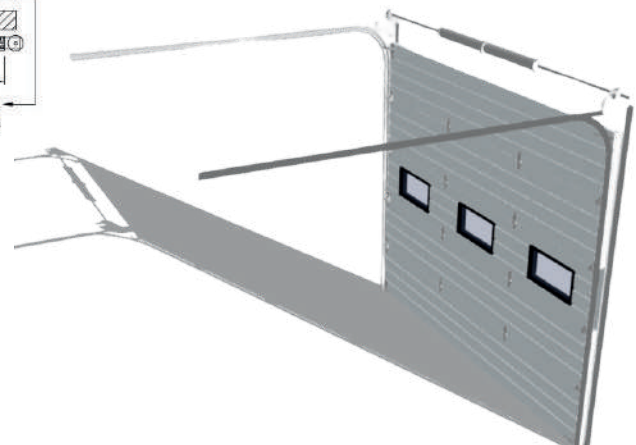


Interior elevation

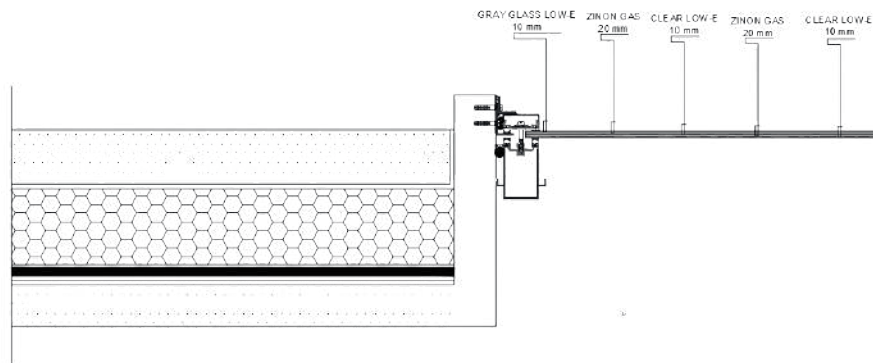


HEADROOM REQUIREMENTS	
DOOR HEIGHT	MINIMUM HEADROOM
7' - 12'	6.12"
12' - 18'	6.12"
18' - 20'	2.50"

* ADD 4" FOR OPERATOR



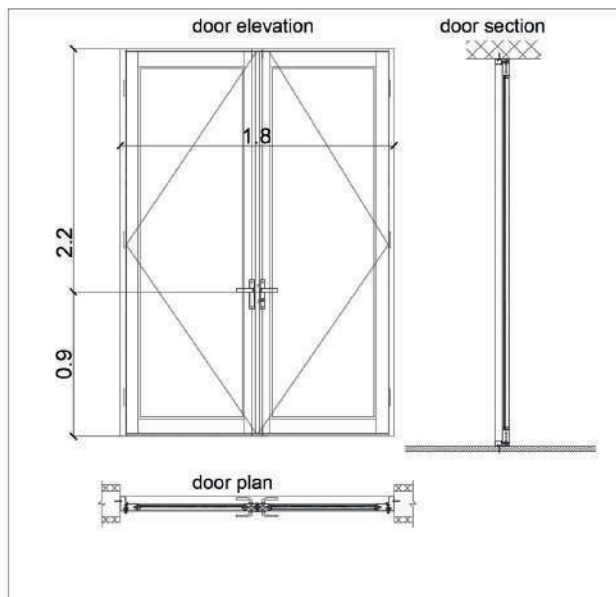
- skylight is Curb Mounted Skylight



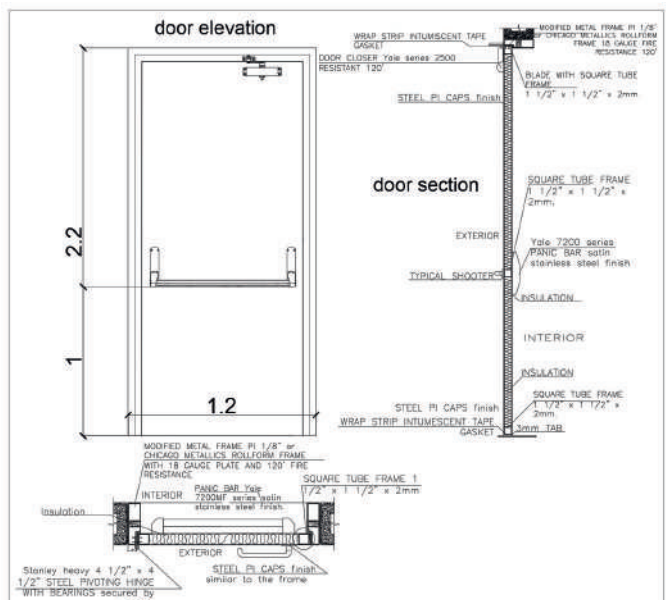
GLASS WALL INSULATION MINIMUM 2.89 W / M² K AS PER SBC. 601

DETAIL OF SKYLIGHT

- Double door glass



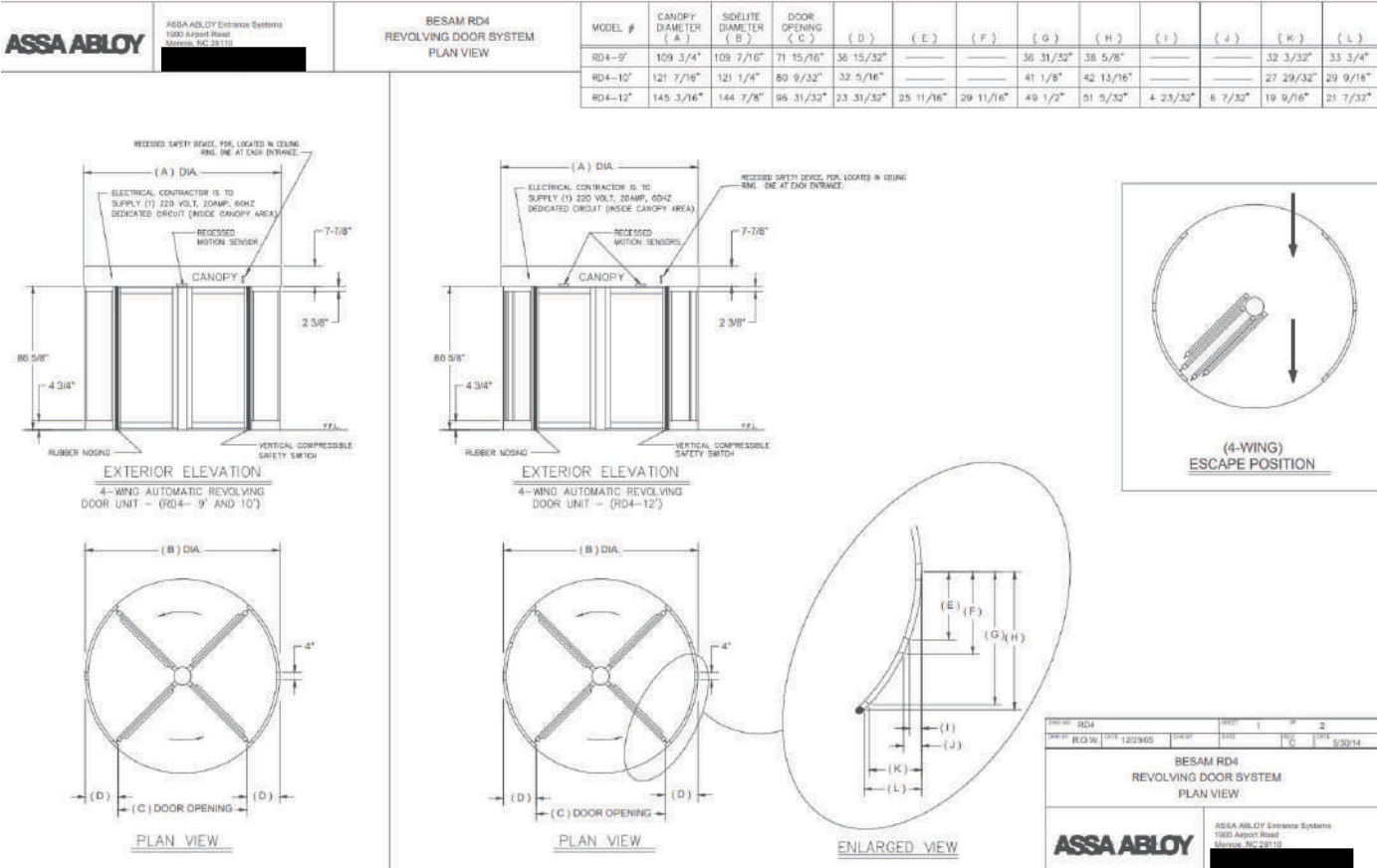
Glass double door elevation



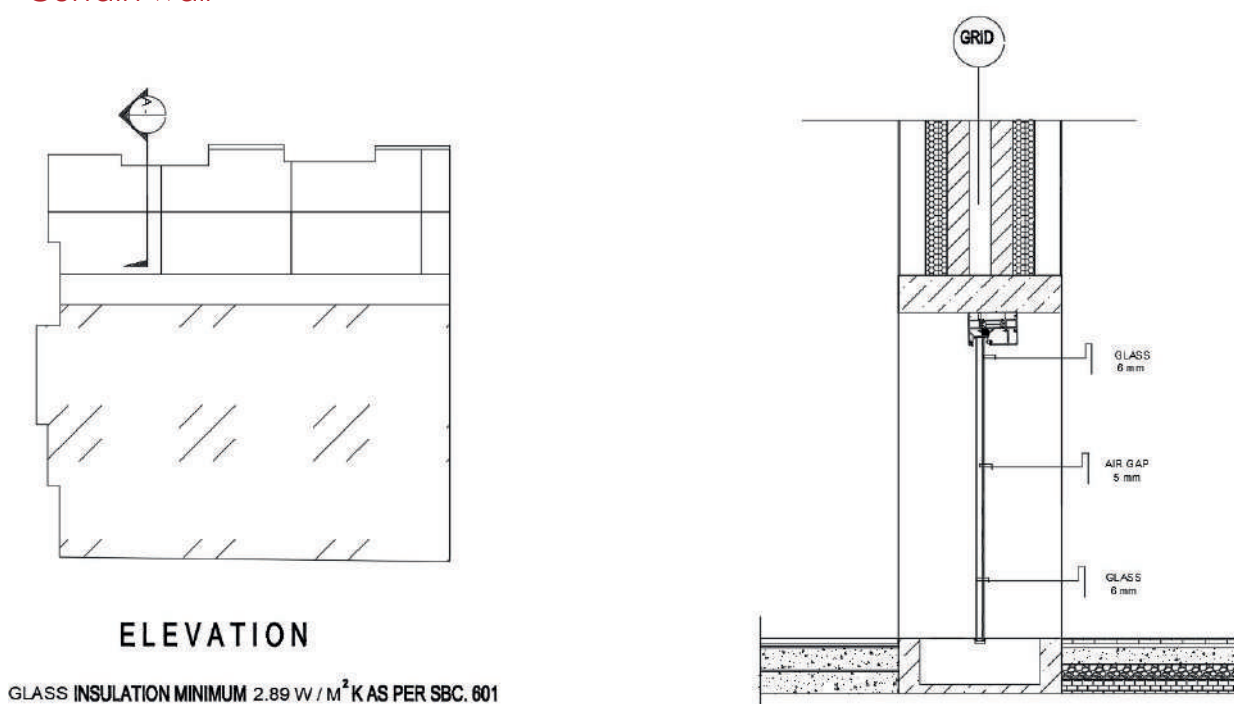
Fire exist single door



• Revolving door

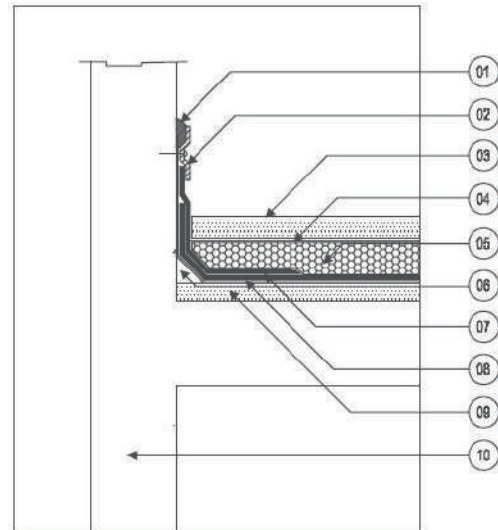
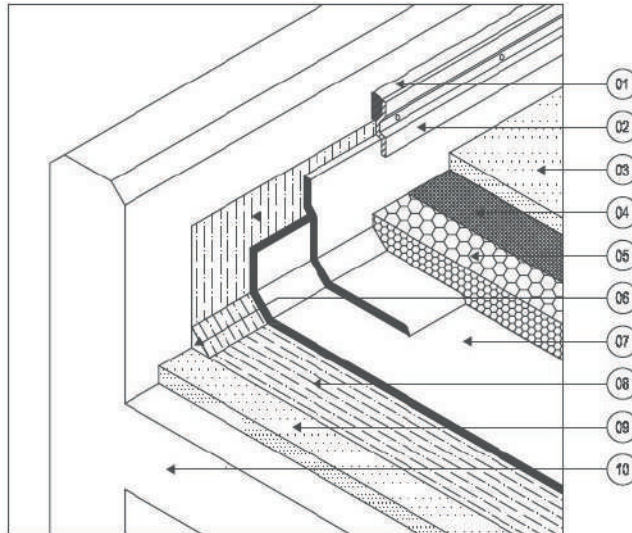


• Curtain wall





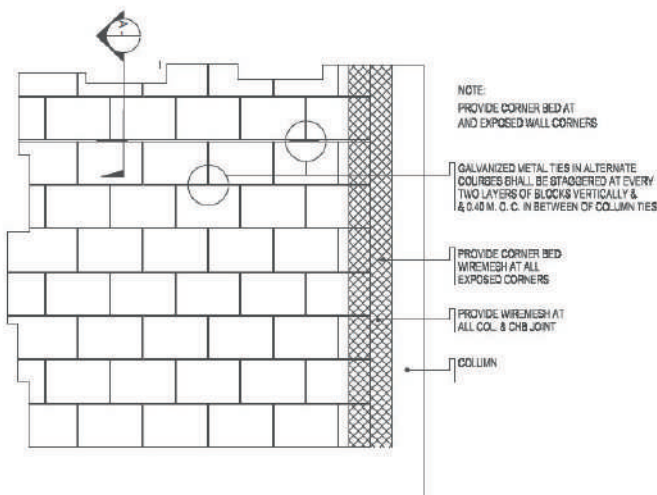
• Roof



INVERTED ROOFING SYSTEM: SAND/CEMENT SCREED FINISH			
SCHEDULE			
NO.	REMARKS	NO.	REMARKS
01	SEALANT	07	VAPOR SEAL 8mm
02	FLASHING	08	BITUMEN 10mm
03	CEMENT SAND RENDER/FINISH 40mm	09	CEMENT SAND RENDER/FINISH 15mm
04	POLYETHYLENE 1mm	10	CONCRETE, CAST 400mm
05	URETHANE / POLYURETHANE 100mm		
06	POLYETHYLENE 1mm		

ROOF INSULATION MINIMUM 0.202 W / M² K AS PER SBC. 601

• Wall

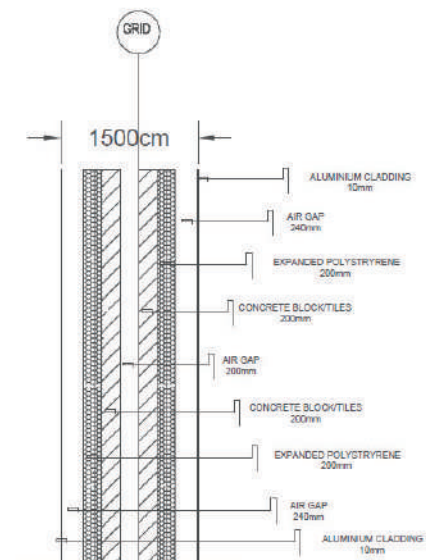


ELEVATION

WALL INSULATION MINIMUM 0.403 W / M² K AS PER SBC. 601

NOT
DETAIL OF INSULATED CAVITY WALL

TO SCALE



NOTE: ANY CHANGES OR MODIFICATION TO THE SPECIFICATIONS OF THE INSULATION MUST MEET THE MIN. SBC 802 REQ.

SECTION-A

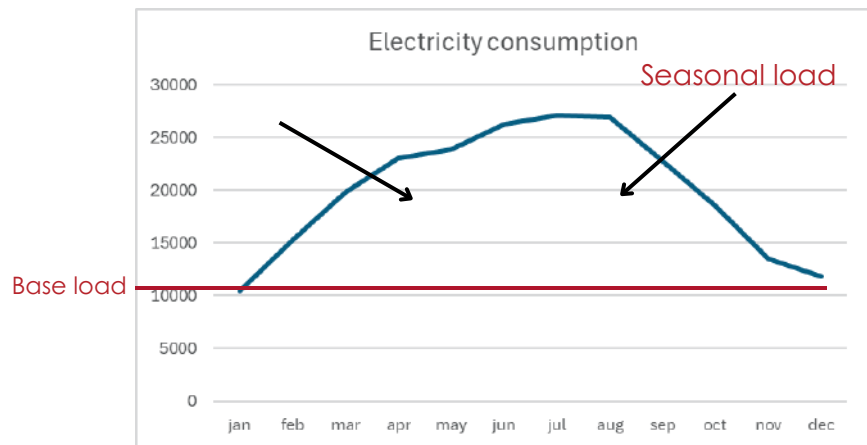
NOT TO SCALE



ENERGY CONSUMPTION INFORMATION

Electricity Bill 2022

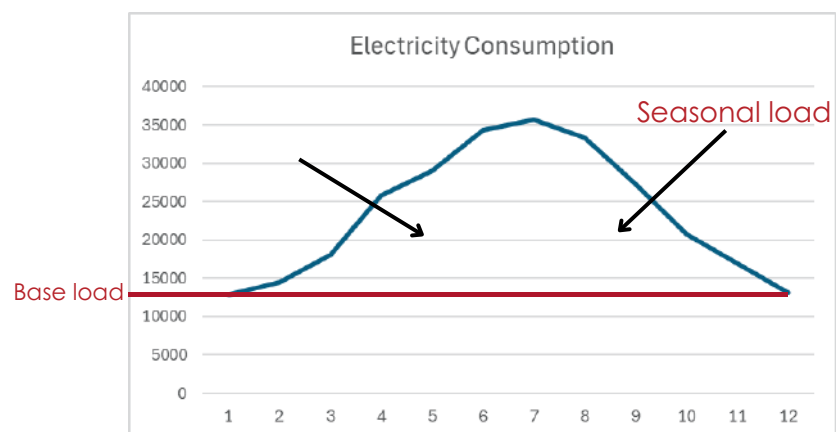
Month	KWh
Jan	10482
Feb	15278
Mar	19834
Apr	23012
May	23849
Jun	26214
Jul	27053
Aug	26910
Sep	22716
Oct	18516
Nov	13503
Dec	11826
Total	239193



- **Annual consumption** = 239,193.0 KWh
- **Monthly base loads** (Lighting+ Equipment)=10,482.0 KWh
- **Annual base loads** (Lighting +Equipment)=10,482.0x12 = 125,784 KWh
- **Annual seasonal load** (HVAC) = 239,193-125,784=113,409KWh

Electricity Bill 2023

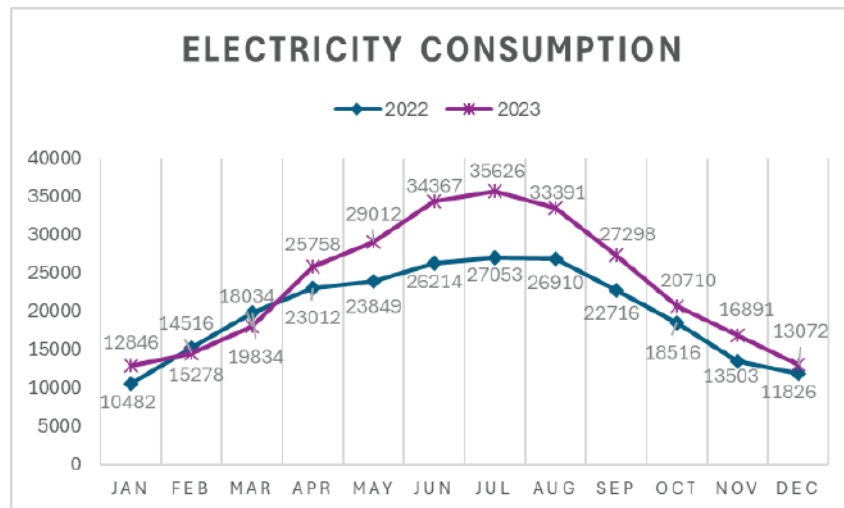
Month	KWh
Jan	12,846
Feb	14,516
Mar	18,034
Apr	25,758
May	29,012
Jun	34,367
Jul	35,626
Aug	33,391
Sep	27,298
Oct	20,710
Nov	16,891
Dec	13,072
Total	281521



- **Annual consumption** =281,521.0 KWh
- **Monthly base loads** (Lighting + Equipment) = 12,846 KWh
- **Annual base loads** (Lighting + Equipment) = 12,846x12=154,152 KWh
- **Annual seasonal load** (HVAC) = 127,369 KWh

ENERGY CONSUMPTION INFORMATION

Electricity consumption between two years :-



Energy use index (EUI)

2022

Total building area: Ground floor (1234) + Mezzanine(525) = 1759

Total KWh: 239,193 KWh

EUI= Total KWh/Total building area

EUI = 239,193/1759 = 135.98 KWh/m²/year

Causes:

- Based on Porsche analysis due to **global crisis** the car market dropped dramatically though that the inflation got higher.
- Employee number **dropped** so the electricity bill is directly proportional
- **Online purchasing** was booming reducing the visitors number to the showroom
- Due to **part shortage** the new car market dropped
- The below figures shows that the **average Temp & Humidity were lower**

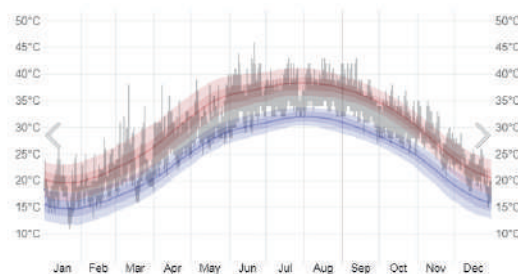


Fig1. 2022 Temp diagram

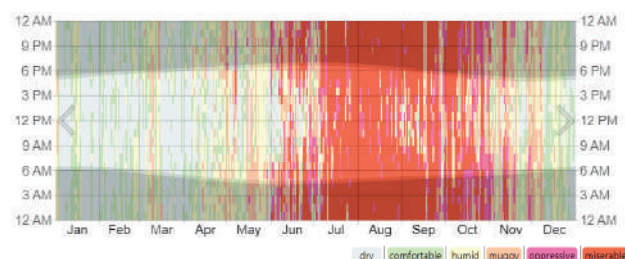


Fig2. 2022 Humidity diagram



2023

Total building area: Ground floor (1234) + Mezzanine(525) = 1759

Total KWh: 281,521 KWh

EUI= Total KWh/Total building area

EUI = 281,521/1759 = 160.04 KWh/m²/year

Causes:

- The below figures shows that the **average Temp & Humidity were Higher.**

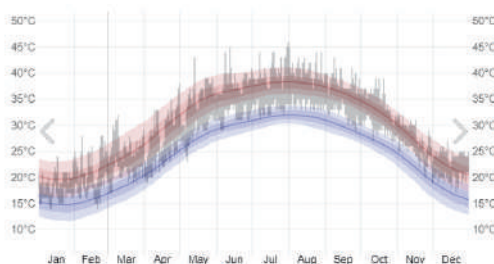


Fig1. 2023 Temp diagram

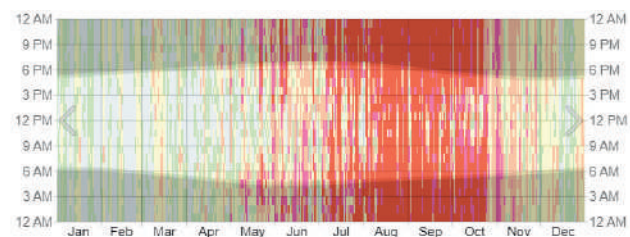


Fig2. 2023 Humidity diagram

- The **Facility Manager Quoted**: At the end of 2023 we found an issues with the HVAC system, The first issue was leakage in the ducts. **Secondly** two of the Package units weren't up to their performance.

- The number of **employment increased**
- Events took place in the showroom 2023 (**The reveal of the new Taycan**).

BUILDING EVALUATION

- 1- There are two open showrooms both triple height that take most the space on the ground floor with a north-east glazing, leading to an increase in cooling load and the intensity of lighting.
- 2- Large skylight starting from the north-east façade climbing up to the roof which increases solar gain.
- 3-Thick wall about 1.5m which delays heat conduction.
- 4- The glass in the north-east façade is about 3m tall with the glass retracted 0.75m back, meaning it needs vertical shading devices to reduce solar gain.
- 5- 90% of the windows are fixed, meaning low infiltration.
- 6- Spotlight LED, due to the function and user needs of a showroom to increase marketing and sales.
- 7- Revolving door is used as entrance, leading to a reduction in infiltration.
- 8-packaged units on the roof, are poor choice of a HVAC system with high running and maintenance cost.



BUILDING EVALUATION

Evaluation form:-

Project Name: Porsche showroom

Date: May 7 2024

Status Code Legend

- Very good =4 POINTS.
- Low/ needs improving =2 POINTS.
- Average =3 POINTS
- Non-existent =1 POINT.

External wall	NORTH-EAST FAÇADE EVALUATION: ● Thermal bridging (2) ● Insulation quality (3) ● Infiltration (1) ● Cracks (4) ● Layers (3) ● Finish exterior colour and materials (3) ● Finish interior colour and materials (3) □ ● TOTAL 19/28 POINTS NORTH-WEST FAÇADE EVALUATION: ● Thermal bridging (3) ● Insulation quality (4) ● Infiltration (4) ● Cracks (4) ● Layers (3) ● Finish exterior colour and materials (3) ● Finish interior colour and materials (3) □ ● TOTAL 24/28 POINTS SOUTH-EAST FAÇADE EVALUATION: ● Thermal bridging (2) ● □ Insulation quality (3) ● □ Infiltration (3) ● □ Cracks (4) ● □ Layers (3) ● □ Finish exterior colour and materials (3) ● □ Finish interior colour and materials (3) □ ● TOTAL 21/28 POINTS
Windows	● Framing material (4) ● Window glazing insulation level (2) ● Window weatherstripping effectiveness (3) ● Window weatherstripping condition (4) ● Window weatherstripping maintenance (4) ● Window tint insulation (1) ● Solar heat gain coefficient value (2) ● Window treatment effectiveness (blinds, curtains, shading, etc) (3) ● Low-e coating level (1) ● Air leakage management (3) ● Age of window (3) □ ● TOTAL 30/44 POINTS



BUILDING EVALUATION

Evaluation form:-

Project Name: Porsche showroom

Date: May 7 2024

Status Code Legend

- Very good =4 POINTS.
- Low/ needs improving =2 POINTS.
- Average =3 POINTS
- Non-existent =1 POINT.

Lighting	● Light source efficiency (4) ● Fixture condition (3) ● Lighting controls (4) ● Daylight integration (4) ● Lighting levels and uniformity (4) ● Lighting quality (2) ● Lighting maintenance (2) ● TOTAL 23/28
Roof	● Roof insulation level (1) Based on now SBC ● Colour of roof insulation level (3) ● Age and condition of roof (2) ● Moisture leak control (2) ● Reflective coating effectiveness (1) ● Roof maintenance plan (2) ● TOTAL 10/24
HVAC	● Cooling capacity (2) ● Energy efficiency rating (2) ● Installation and fit (1) ● Airflow and distribution (1) ● Noise level (4) ● Air quality features (2) ● User-friendly controls (2) ● Durability and reliability (1) ● Energy efficiency measures (2) ● Maintenance requirements (1) ● TOTAL 18/40
Door	● Door material insulation level (1) ● Weather sealing level (1) ● Glazing insulation level in door (2) ● Air leakage management level (1) ● Blower door mechanism efficiency (1) ● Self-closing mechanism efficiency (2) ● Condition of weather stripping (2) ● Maintenance for weatherstripping on doors (2) ● TOTAL 12/32



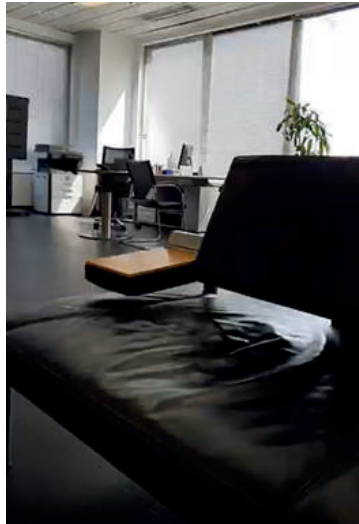
WALK-THROUGH AUDIT

Picture:

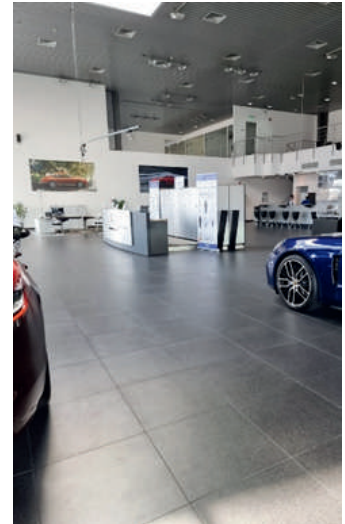
Entrance



Employee offices



Reception



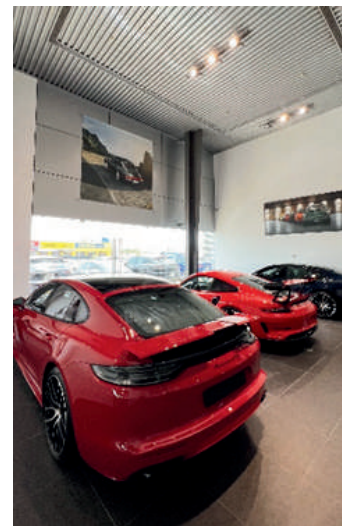
Lounge



Coffee corner



Used car



South-west



North-East





WALK-THROUGH AUDIT

Measurements:



Showroom:

Analysis

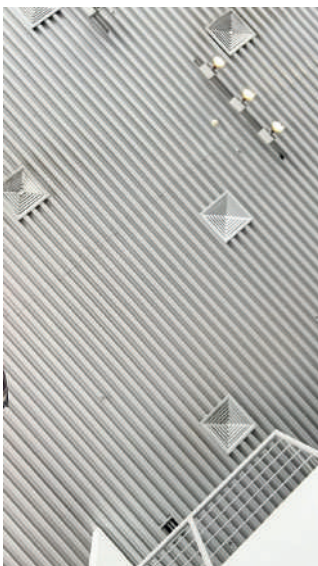


CURTAIN WALL, CLADDING & SKYLIGHT

After asking the facilities manager about curtain wall & skylight we were informed that the type of curtain wall used is (High-Performance Double Glass) & skylight is (Curb Mounted Skylight). Type of cladding used is metal panel .

HVAC

System used is package system, it's placed in the roof of the building. The ducts run from package to supply diffuser and grill. Due to the building being a car showroom, there must be a CO2 detector, and once CO2 exceeds the maximum allowed value 20ppm, the HVAC will operate at a higher speed to clean the air. This was integrated in the Porsche showroom.



WALK-THROUGH AUDIT

Measurements:



Showroom:

Analysis

COMMENTS

- The lights are always on and computers, this can increase your electric bill.
- A malfunctioning revolving door causes HVAC to work harder, due to infiltration, resulting in increased energy use.
- An improperly programmed HVAC system, particularly one that runs continuously lead to overspending.



AIR TEMPERATURE

According to the standard, temperature should be between 20C and 26C. measure result was 18C it's cold



HUMIDITY

According to the standard, humidity should be between 40 -60%, measure result was 32% it's not acceptable



According to the standards, lux should be 1000 – 1500, measure result was 737 it's not acceptable

LIGHTING

WALK-THROUGH AUDIT

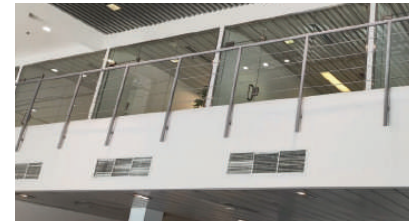
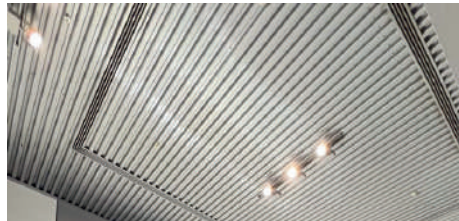
10

Measurements:



Showroom for used cars:

Analysis



HVAC

System used is **package system**, it's placed in the **roof** of the building. The ducts run from package to supply **grill**. Due to the building being a car showroom, there must be a CO detector, and once CO exceeds the maximum allowed value **20ppm**, the hvac will operate at a higher speed to clean the air. This was integrated in the Porsche used car.

COMMENTS

- Lights (Always On): Constant lighting use significantly increases electricity consumption.
- Door (Air Infiltration): A faulty door allows uncontrolled outside air to enter, forcing the HVAC to constantly adjust, wasting energy.
- HVAC (Constant Operation): Constant operation, especially due to door issues, leads to high energy use.

CURTAIN WALL & GYPSUM BOARD

After asking the facilities manager about curtain wall it is clear that the type of curtain wall is **(High-Performance Double Glass)** and **Fire-Resistant Gypsum Board (Type X)**



AIR TEMPERATURE & FLOW

According to the standard, temperature should be between 20C and 26C. measure result was 19C it's cold

HUMIDITY

According to the standard, humidity should be between 40 -60%, measure result was 32% it's not acceptable

LIGHTING

According to the standards, lux should be 1000 – 1500, measure result was 673 it's not acceptable

WALK-THROUGH AUDIT

Measurements:



Lounge:



GYPSMBOARD

Fire-Resistant Gypsum Board offers thermal resistance, it's providing more, potentially reducing HVAC system workload and energy consumption by keeping conditioned air inside the room.



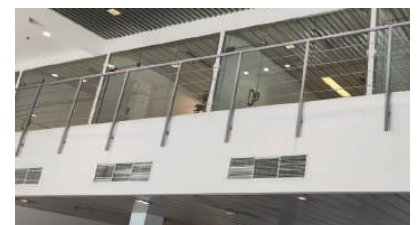
AIR TEMPERATURE & FLOW

According to the standard, temperature should be between 20C and 26C. measure result was 20C it's acceptable



HVAC

System used is package system, it's placed in the roof of the building. The ducts run from package to grill



Analysis

COMMENTS

- Lights (Always On): Constant illumination unnecessarily drains energy.
- TV (Always On): Continuous TV operation adds to the energy.
- HVAC (Constant Operation): Constant Operation of HVAC use, likely exacerbated by constant heat gain from electronics, drives up energy costs.



LIGHTING

According to the standards, lux should be 1000 – 1500, measure result was 673 it's not acceptable



HUMIDITY

According to the standard, humidity should be between 40 -60%, measure result was 32% it's not acceptable

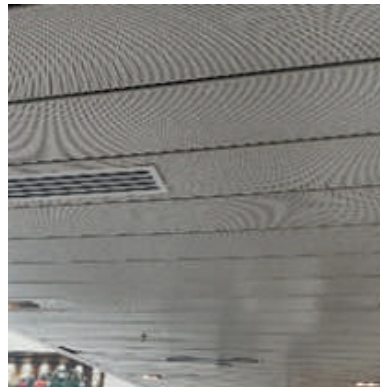
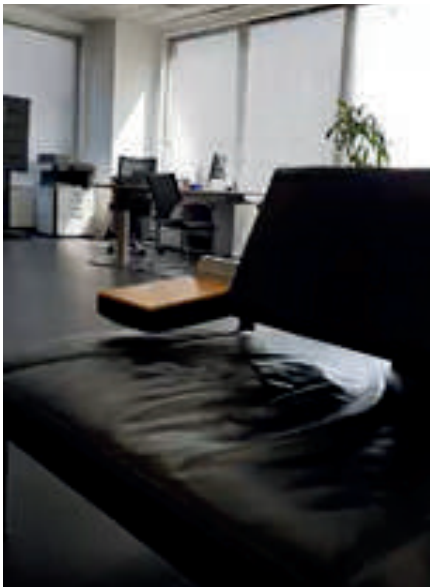
WALK-THROUGH AUDIT

Measurements:



Employee offices

Analysis



HVAC

System used is **package system**, it's placed in the **roof** of the building. The ducts run from package to **grill**

GYPSMBOARD

Fire-Resistant Gypsum Board offers thermal resistance, it's providing more, potentially reducing HVAC system workload and energy consumption by keeping conditioned air inside the room.

COMMENTS

- Always-on Lights & Electronics: Continuously running lights, computers, and TVs drain energy.
- Constantly Running HVAC: If not programmed well, the HVAC can run even when offices are empty, wasting power.



AIR TEMPERATURE & FLOW

According to the standard, temperature should be between 20C and 26C. measure result was **20C** it's **acceptable**



HUMIDITY

According to the standard, humidity should be between 40 -60%, measure result was 32% it's not acceptable



LIGHTING

According to the standards, lux should be **1000 – 1500**, measure result was **673** it's **not acceptable**

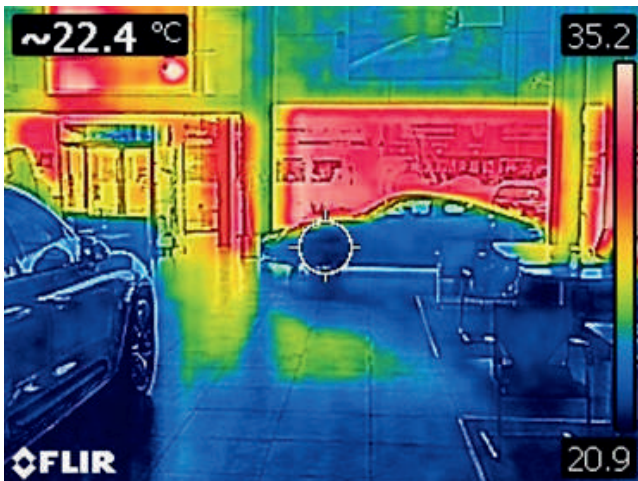
WALK-THROUGH AUDIT

Measurements:



Showroom

Infrared



The image shows that the glass isn't a thermal insulated. There is infiltration coming from revolving door (not working) and curtain wall, because of poor maintenance.

Used car showroom



The image shows that the glass isn't a thermal insulated. There is infiltration coming from the curtain wall because of poor maintenance.



South west



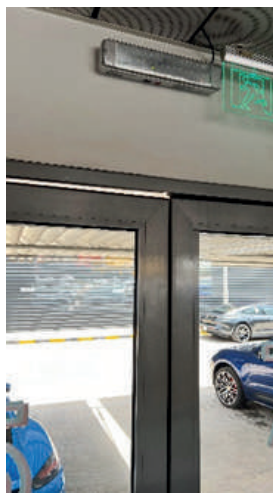
High levels of heat is entering the building due to the garage, and on delivery times, the garage is opened for a long time causing high levels of heat gain

WALK-THROUGH AUDIT

Measurements:

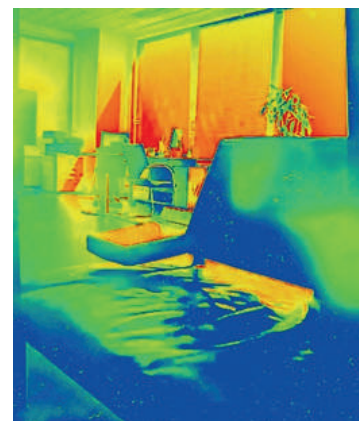
All doors infiltration

Infrared



High heat conductivity in the frame of the door. And high infiltration due to poor maintenance and the door is not closed properly most of the time.

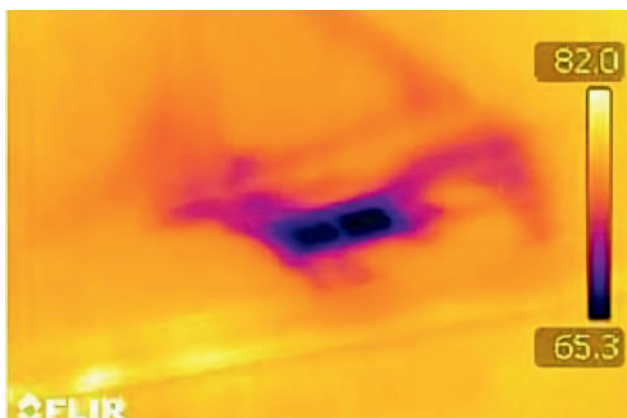
Employee offices



High heat conductivity in window glass

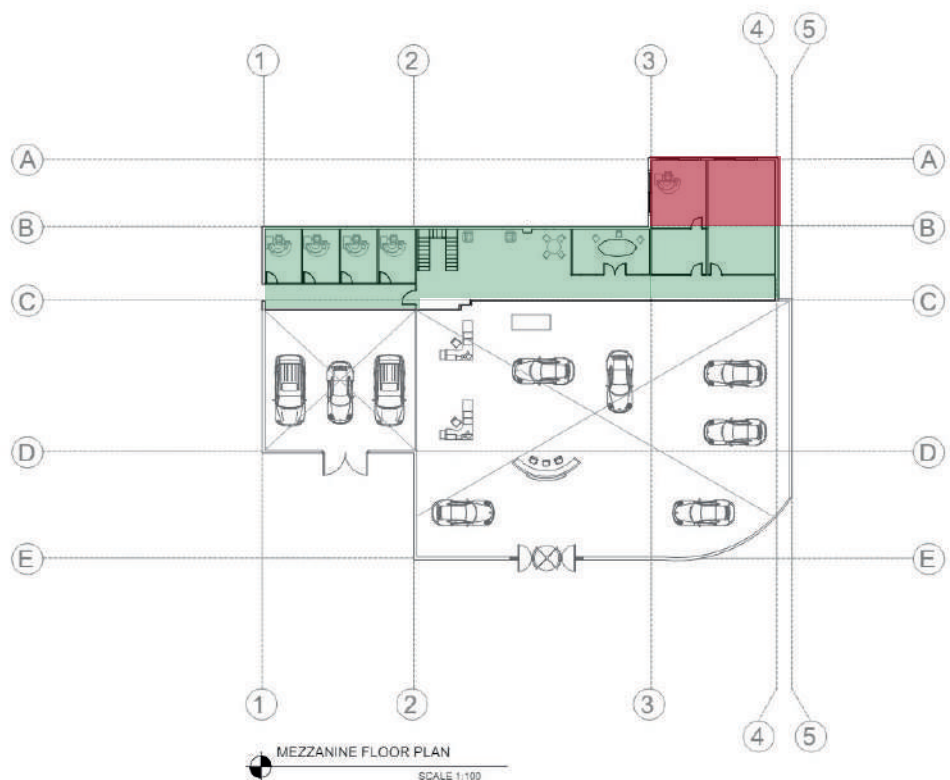
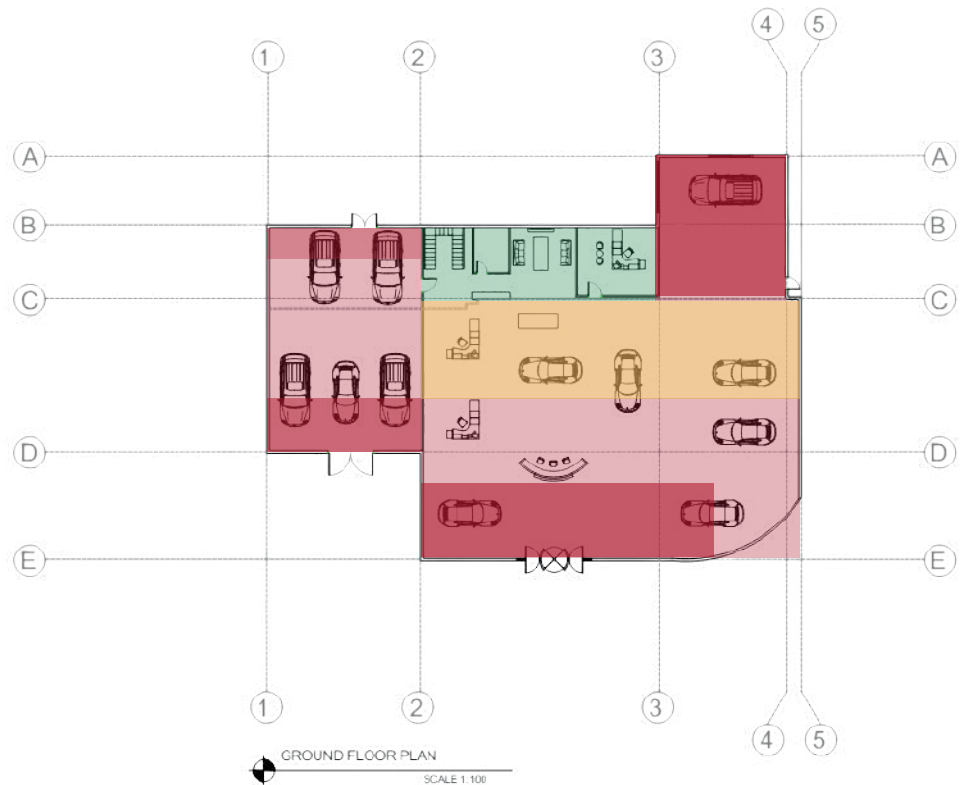
Refrigerant Leaks::

Duct leakage in HVAC system



The cooler duct section suggests conditioned air leakage from the duct into the surrounding area. This can reduce the efficiency of the HVAC system as it has to work harder to maintain the desired temperature.

DEFINE AREAS OF HIGH ENERGY USED



RECOMMENDATION

1- Doors



DUE TO BACK DOOR NOT CLOSING PROPERLY

- Installing the automatic sliding door with an air curtain, this combo will result in 62% reduction of heat

DUE TO MAIN ENTRANCE REVOLVING DOOR NOT OPERATING AUTOMATICALLY

- Fix the door and make sure all the sealants are in perfect condition to minimize infiltration

DUE TO THE INFILTRATION OF BACK GARAGE :

Install double garage to reduce infiltration

2- Windows

DUE TO ORIENTATION OF THE BUILDING (NORTH-EAST)

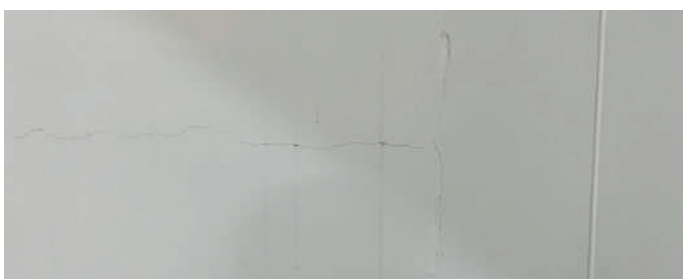
- We need shading devices (vertical).
- Change glazing type (low-e coating, high efficiency double glass , coloured glass).
- Higher quality weather stripping



3- Walls



Due to the steel column structural place in the façade causing thermal bridging



- Column cover with insulation and gypsum board
- Find a solution for the cracks in the walls

RECOMMENDATION

4- Roof



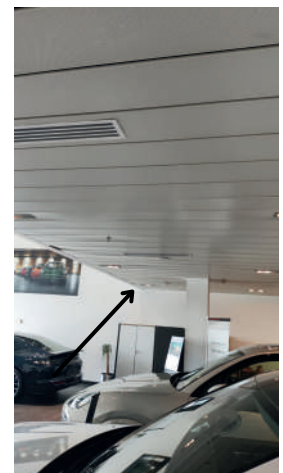
Due to the roof improvement and minimize heat observation heat observation

- paint the roof with lighter colours
- use PV system to use it during the break time to save electricity
- double roof system

5- Lighting

Due to the function of the building and the need to focus on the cars:

- Use more efficiency spotlight led
- putting the focal point under the sky light to use the natural lighting



6- A/C system



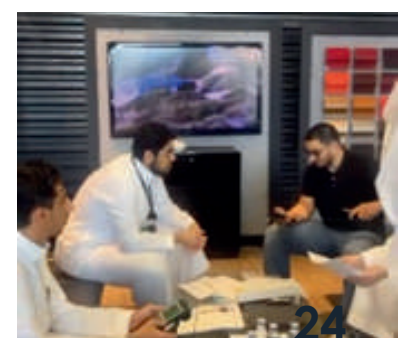
Due to the air packaged units are all exposed to the sun rays

- Use shading above them to help with the performance and help maintain the system

7- Equipment & Load Finding

Due to the tv is showing the advertising clips while no one there

- Installing a motion detector so that when no one is around, it will automatically turn off and on when someone walks by.





RECOMMENDATION

Window	• Change glazing type (low-e coating, high efficiency Triple glass, coloured glass). • higher quality weather stripping.
Walls	• Increase the insulation • Close the air gaps • Weather the cladding infiltration fix the wall cracks.
Roof	• Paint the roof with lighter colours. • Use PV-system to use it during the break time to save electricity. • Double roof system.
Lighting	• Turn off the light during the break time. • Use a type of lighting that suitable for the show room. • Increase the number of lighting in the show room • Upgrade the old led light (20w) with a (40w-60w). • Putting the focal point under the sky light to use the natural lighting.
A/C system	• We can put shade above them to help with the performance and help maintain the system. • Change the system to concealed. • Change the ductwork and the leakage problems
equipment loads findings	• Installing a motion detector so that when no one is around, it will automatically turn off and on when someone walks by. • all electric devices must be unplugged during off hours.



CASE 1

CURRENT SITUATION

ACTIVITY INPUT

Porsche Dealership	
Template	PORSCHE DEALERSHIP
Sector	B1 Offices and Workshop businesses
Zone multiplier	1
☒ Include zone in thermal calculations	
☒ Include zone in Radiance daylighting calculations	
Floor Areas and Volumes	
Occupancy	
Occupancy density (people/m2)	0.1500
Schedule	PORSCHE SCHEDULE
Metabolic	
Activity	Standing/walking
Factor (Men=1.00, Women=0.85, Children=...	0.95
CO2 generation rate (m3/s-W)	0.000000362
Clothing	
Clothing schedule definition	1-Generic summer and winter clothing
Winter clothing (clo)	1.00
Summer clothing (clo)	0.70
Comfort Radiant Temperature Weighting	
Contaminant Generation and Removal	
Holidays	
☒ Holidays	
Holidays per year	12
Holiday schedule	Copy of SAUDI HOLIDAYS
DHW	
Environmental Control	
Heating Setpoint Temperatures	
Heating (°C)	20.0
Heating set back (°C)	18.0
Cooling Setpoint Temperatures	
Cooling (°C)	24.0
Cooling set back (°C)	28.0
Humidity Control	
RH Humidification setpoint (%)	40.0
RH Dehumidification setpoint (%)	60.0

LIGHTING INPUT

Lighting	
Target Illuminance (lux)	737
Default display lighting density (W/m2)	9

COMPUTER & OFFICE EQUIPMENT

Computers	
☒ On	
Power density (W/m2)	1.00
Schedule	PORSCHE SCHEDULE
Radiant fraction	0.200
Office Equipment	
☒ On	
Power density (W/m2)	8.00
Schedule	Office_CarPark_Equip
Radiant fraction	0.200



CASE 1

AIRTIGHTNESS

Airtightness	
☒ Model infiltration	
Constant rate (ac/h)	0.900

WALL SECTION DETAIL

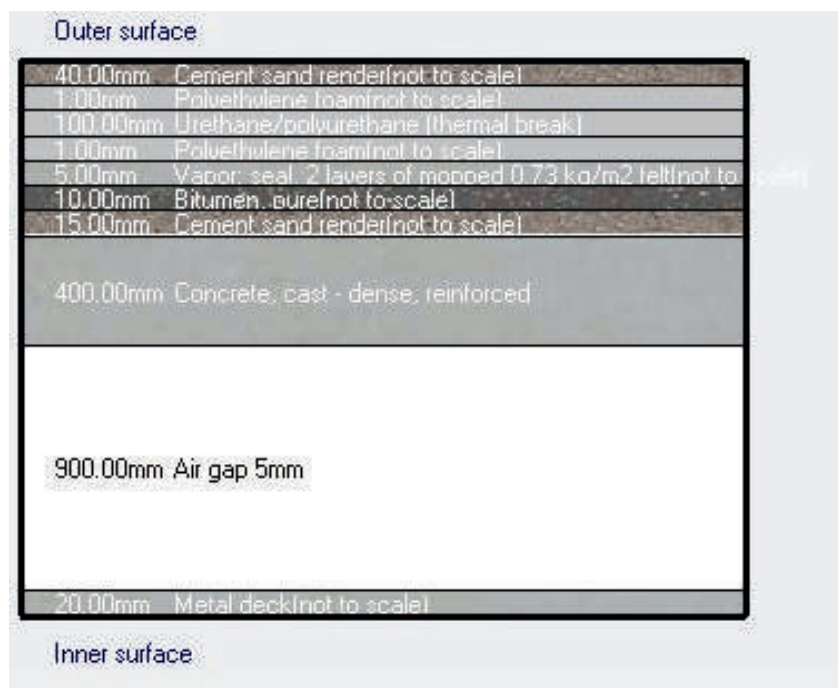
Outer surface	
10.00mm	Aluminium(not to scale)
240.00mm	Air gap ≥ 25 mm
200.00mm	EPS Expanded Polystyrene (Lightweight)
200.00mm	Concrete blocks/tiles - block, aerated
200.00mm	Air gap 10mm
200.00mm	Concrete blocks/tiles - block, aerated
200.00mm	EPS Expanded Polystyrene (Lightweight)
240.00mm	Air gap 10mm
10.00mm	Aluminium(not to scale)
Inner surface	

CONSTRUCTION DEATILS LAYERS

Edit construction - PORSCHE WALL	
Constructions	
Layers	Surface properties
Image	Calculated
Cost	Condensation analysis
Inner surface	
Convective heat transfer coefficient (W/m ² -K)	5.846
Radiative heat transfer coefficient (W/m ² -K)	1.847
Surface resistance (m ² -K/W)	0.130
Outer surface	
Convective heat transfer coefficient (W/m ² -K)	23.290
Radiative heat transfer coefficient (W/m ² -K)	1.710
Surface resistance (m ² -K/W)	0.040
No Bridging	
U-Value surface to surface (W/m ² -K)	0.092
R-Value (m ² -K/W)	11.012
U-Value (W/m²-K)	0.091
With Bridging (BS EN ISO 6946)	
Thickness (m)	1.5000
Km - Internal heat capacity (KJ/m ² -K)	24.6400
Upper resistance limit (m ² -K/W)	11.012
Lower resistance limit (m ² -K/W)	11.012
U-Value surface to surface (W/m ² -K)	0.092
R-Value (m ² -K/W)	11.012
U-Value (W/m²-K)	0.091



CASE 1



ROOF SECTION DETAIL

Edit construction - PORSCHE ROOF

Constructions	
Layers	Surface properties
Image	Calculated
Cost	Condensation analysis
Inner surface	
Convective heat transfer coefficient (W/m ² -K)	4.460
Radiative heat transfer coefficient (W/m ² -K)	5.540
Surface resistance (m ² -K/W)	0.100
Outer surface	
Convective heat transfer coefficient (W/m ² -K)	19.870
Radiative heat transfer coefficient (W/m ² -K)	5.130
Surface resistance (m ² -K/W)	0.040
No Bridging	
U-Value surface to surface (W/m ² -K)	0.861
R-Value (m ² -K/W)	1.301
U-Value (W/m²-K)	0.769
With Bridging (BS EN ISO 6946)	
Thickness (m)	1.4920
Km - Internal heat capacity (KJ/m ² -K)	78.2400
Upper resistance limit (m ² -K/W)	1.301
Lower resistance limit (m ² -K/W)	1.301
U-Value surface to surface (W/m ² -K)	0.861
R-Value (m ² -K/W)	1.301
U-Value (W/m²-K)	0.769

CONSTRUCTION DEATILS LAYERS



CASE 1

Calculated Values	
Total solar transmission (SHGC)	0.503
Direct solar transmission	0.373
Light transmission	0.505
U-value (ISO 10292/ EN 673) (W/m2-K)	3.239
U-Value (W/m2-K)	3.094

CALCULATED VALUE SHGC

Lighting Template

Template LED

General Lighting

☒ On

Normalised power density (W/m2-100 lux) 2.5000

Schedule PORSCHE SCHEDULE

Luminaire type 1-Suspended

Return air fraction 0.000

Radiant fraction 0.420

Visible fraction 0.180

Convective fraction 0.400

LIGHT TYPE DATA

HVAC Template

Template Packaged DX

Mechanical Ventilation

☒ On

Outside air definition method 4-Min fresh air (Sum per person + per area)

Operation

Schedule PORSCHE REAL HVAC SCHEDULE

Economiser (Free Cooling) >>

Heat Recovery >>

Auxiliary Energy >>

Heating >>

Cooling

☒ Cooled

Cooling system Default

Fuel 1-Electricity from grid

Cooling system seasonal CoP 2

Supply Air Condition

Minimum supply air temperature (°C) 12.00

Minimum supply air humidity ratio (g/g) 0.0077

Cooling limit type 3-Limit flow rate and capacity

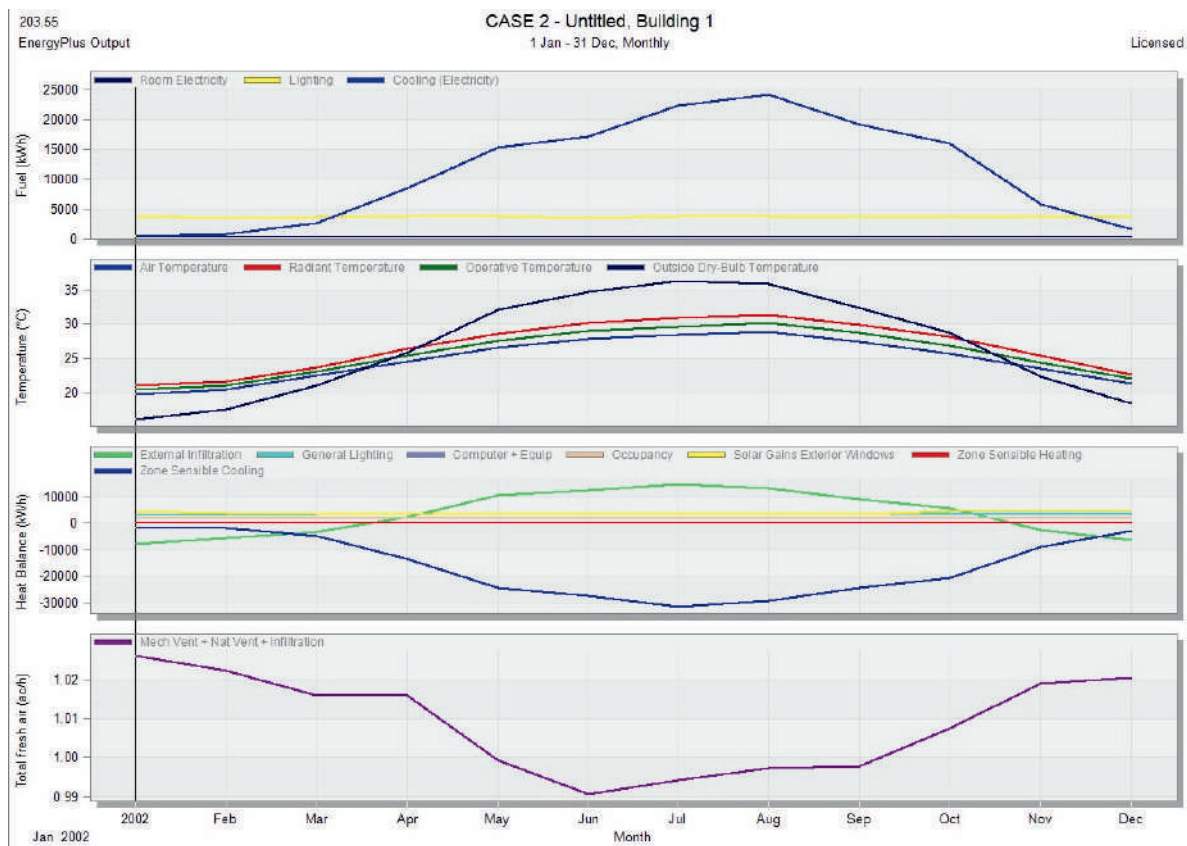
Operation

Schedule PORSCHE REAL HVAC SCHEDULE

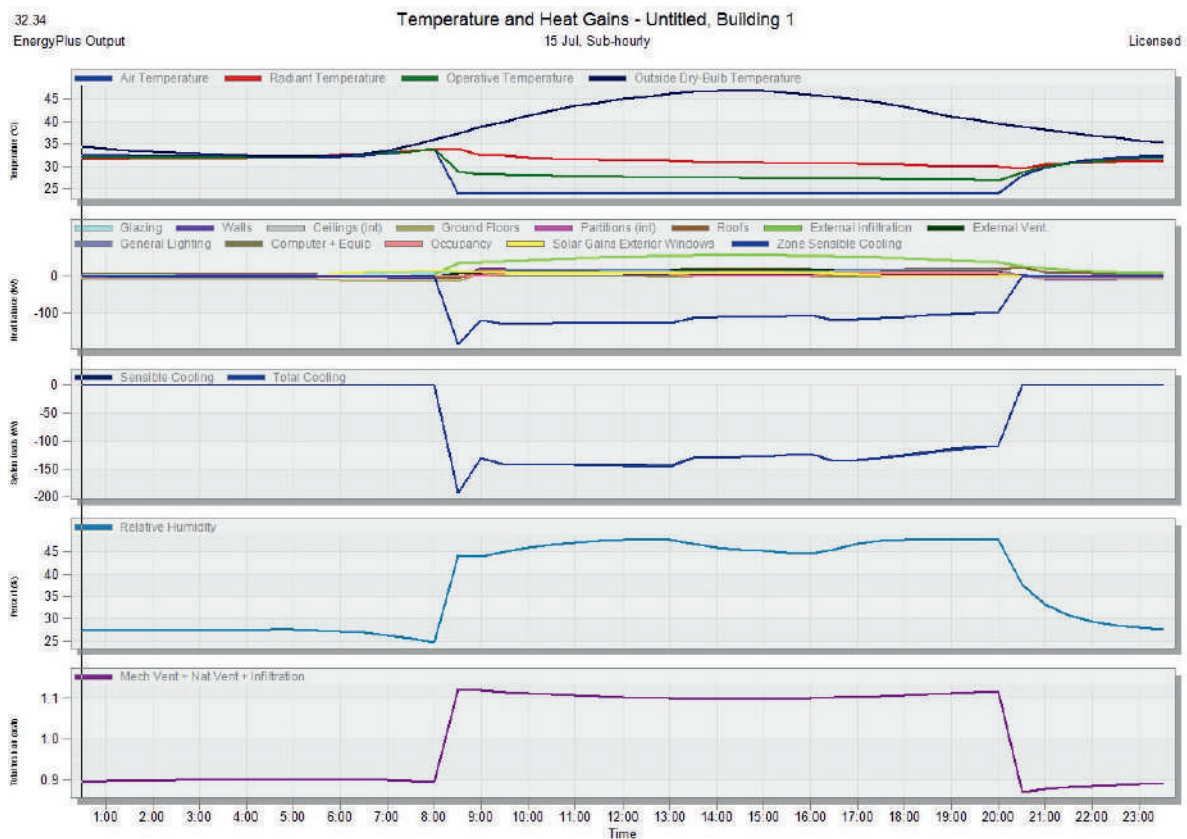
HVAC DATA



CASE 1



SIMULATION ANALYSIS



COOLING DESIGN ANALYSIS



CASE 1

Profiles							
Month	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Jan	8:00-20:00	8:00-20:00	8:00-20:00	8:00-20:00	Off	8:00-21:00	8:00-20:00
Feb	8:00-20:00	8:00-20:00	8:00-20:00	8:00-20:00	Off	8:00-21:00	8:00-20:00
Mar	8:00-20:00	8:00-20:00	8:00-20:00	8:00-20:00	Off	8:00-21:00	8:00-20:00
Apr	8:00-20:00	8:00-20:00	8:00-20:00	8:00-20:00	Off	8:00-21:00	8:00-20:00
May	8:00-20:00	8:00-20:00	8:00-20:00	8:00-20:00	Off	8:00-21:00	8:00-20:00
Jun	8:00-20:00	8:00-20:00	8:00-20:00	8:00-20:00	Off	8:00-21:00	8:00-20:00
Jul	8:00-20:00	8:00-20:00	8:00-20:00	8:00-20:00	Off	8:00-21:00	8:00-20:00
Aug	8:00-20:00	8:00-20:00	8:00-20:00	8:00-20:00	Off	8:00-21:00	8:00-20:00
Sep	8:00-20:00	8:00-20:00	8:00-20:00	8:00-20:00	Off	8:00-21:00	8:00-20:00
Oct	8:00-20:00	8:00-20:00	8:00-20:00	8:00-20:00	Off	8:00-21:00	8:00-20:00
Nov	8:00-20:00	8:00-20:00	8:00-20:00	8:00-20:00	Off	8:00-21:00	8:00-20:00
Dec	8:00-20:00	8:00-20:00	8:00-20:00	8:00-20:00	Off	8:00-21:00	8:00-20:00

FIRST CASE REAL HVAC SCHEDULE

Profiles							
Month	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Jan	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
Feb	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
Mar	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
Apr	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
May	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
Jun	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
Jul	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
Aug	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
Sep	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
Oct	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
Nov	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
Dec	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00

WORKING HOURS SCHEDULE

Site and Source Energy

	Total Energy [kWh]	Energy Per Total Building Area [kWh/m ²]	Energy Per Conditioned Building Area [kWh/m ²]
Total Site Energy	314045.37	364.10	364.10
Net Site Energy	314045.37	364.10	364.10
Total Source Energy	429407.73	497.85	497.85
Net Source Energy	429407.73	497.85	497.85

Utility Use Per Total Floor Area

	Electricity Intensity [kWh/m ²]	Natural Gas Intensity [kWh/m ²]	Additional Fuel Intensity [kWh/m ²]	District Cooling Intensity [kWh/m ²]	District Heating Intensity [kWh/m ²]	Water Intensity [m ³ /m ²]
Lighting	50.98	0.00	0.00	0.00	0.00	0.00
HVAC	0.00	0.00	0.00	310.35	0.00	0.00
Other	2.77	0.00	0.00	0.00	0.00	0.00
Total	53.75	0.00	0.00	310.35	0.00	0.00

End Uses

	Electricity [kWh]	Natural Gas [kWh]	Additional Fuel [kWh]	District Cooling [kWh]	District Heating [kWh]	Water [m ³]
Heating	0.00	0.00	0.00	0.00	0.00	0.00
Cooling	0.00	0.00	0.00	267665.81	0.00	0.00
Interior Lighting	43972.97	0.00	0.00	0.00	0.00	0.00
Exterior Lighting	0.00	0.00	0.00	0.00	0.00	0.00
Interior Equipment	2386.59	0.00	0.00	0.00	0.00	0.00
Exterior Equipment	0.00	0.00	0.00	0.00	0.00	0.00
Fans	0.00	0.00	0.00	0.00	0.00	0.00
Pumps	0.00	0.00	0.00	0.00	0.00	0.00
Heat Rejection	0.00	0.00	0.00	0.00	0.00	0.00
Humidification	0.00	0.00	0.00	0.00	0.00	0.00
Heat Recovery	0.00	0.00	0.00	0.00	0.00	0.00
Water Systems	0.00	0.00	0.00	0.00	0.00	0.00
Refrigeration	0.00	0.00	0.00	0.00	0.00	0.00
Generators	0.00	0.00	0.00	0.00	0.00	0.00
Total End Uses	46359.57	0.00	0.00	267665.81	0.00	0.00

Electric Loads Satisfied

	Electricity [kWh]	Percent Electricity [%]
Fuel-Fired Power Generation	0.000	0.00
High Temperature Geothermal*	0.000	0.00
Photovoltaic Power	0.000	0.00
Wind Power	0.000	0.00
Power Conversion	0.000	0.00
Net Decrease in On-Site Storage	0.000	0.00
Total On-Site Electric Sources	0.000	0.00
Electricity Coming From Utility	46359.566	100.00
Surplus Electricity Going To Utility	0.000	0.00
Net Electricity From Utility	46359.566	100.00
Total On-Site and Utility Electric Sources	46359.566	100.00
Total Electricity End Uses	46359.566	100.00

INTERIOR EQUIPMENT AND LIGHTING



CASE 2 & SAUDI BUILDING CODE INPUT

ACTIVITY INPUT

Template		PORSCHE DEALERSHIP
Sector	B1 Offices and Workshop businesses	
Zone multiplier	1	
☒ Include zone in thermal calculations		
☒ Include zone in Radiance daylighting calculations		
Floor Areas and Volumes		
Occupancy		
Occupancy density (people/m ²)	0.1500	
Schedule	PORSCHE SCHEDULE	
Metabolic		
Activity	Standing/walking	
Factor (Men=1.00, Women=0.85, Children=...	0.95	
CO ₂ generation rate (m ³ /s-W)	0.000000362	
Clothing		
Clothing schedule definition	1-Generic summer and winter clothing	
Winter clothing (clo)	1.00	
Summer clothing (clo)	0.70	
Comfort Radiant Temperature Weighting		
Contaminant Generation and Removal		
Holidays		
☒ Holidays		
Holidays per year	12	
Holiday schedule	Copy of SAUDI HOLIDAYS	
DHW		
Environmental Control		
Heating Setpoint Temperatures		
Heating (°C)	20.0	
Heating set back (°C)	18.0	
Cooling Setpoint Temperatures		
Cooling (°C)	24.0	
Cooling set back (°C)	28.0	
Humidity Control		
RH Humidification setpoint (%)	40.0	
RH Dehumidification setpoint (%)	60.0	

LIGHTING INPUT

Lighting	
Target Illuminance (lux)	1200
Default display lighting density (W/m ²)	9

COMPUTER & OFFICE EQUIPMENT

Computers	
☒ On	
Power density (W/m ²)	1.00
Schedule	PORSCHE SCHEDULE
Radiant fraction	0.200
Office Equipment	
☒ On	
Power density (W/m ²)	8.00
Schedule	Office_CarPark_Equip
Radiant fraction	0.200



CASE 2 & SAUDI BUILDING CODE INPUT

AIRTIGHTNESS

Airtightness	
☒ Model infiltration	
Constant rate (ac/h)	0.600
☐ Schedule	On 24/7

WALL SECTION DETAIL

Outer surface
10.00mm Metals - stainless steel, 20/Ni(not to scale)
240.00mm Air gap >=25mm
200.00mm EPS Expanded Polystyrene (Heavyweight)
2.00mm Bitumen, felt/sheet(not to scale)
200.00mm Concrete blocks/tiles - block, perlite-filled, mediumweight
200.00mm EPS Expanded Polystyrene (Heavyweight)
200.00mm Concrete blocks/tiles - block, perlite-filled, mediumweight
240.00mm EPS Expanded Polystyrene (Heavyweight)
240.00mm Air gap 10mm
10.00mm Metals - stainless steel, 20/Ni(not to scale)

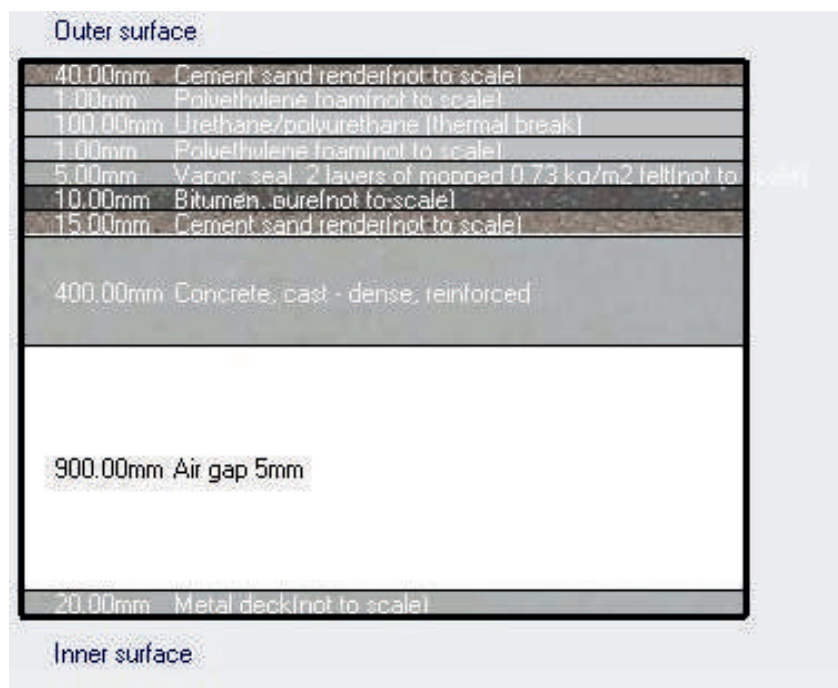
Inner surface

CONSTRUCTION DETAILS LAYERS

Edit construction - PORSCHE WALL

Outer surface	
Convective heat transfer coefficient (W/m ² -K)	2.152
Radiative heat transfer coefficient (W/m ² -K)	5.540
Surface resistance (m ² -K/W)	0.130
Inner surface	
Convective heat transfer coefficient (W/m ² -K)	19.870
Radiative heat transfer coefficient (W/m ² -K)	5.130
Surface resistance (m ² -K/W)	0.040
Bridging	
U-Value surface to surface (W/m ² -K)	0.048
R-Value (m ² -K/W)	20.796
U-Value (W/m²-K)	0.048
Thermal Bridging (BS EN ISO 6946)	
Thickness (m)	1.5420
Km - Internal heat capacity (kJ/m ² -K)	38.4000
Upper resistance limit (m ² -K/W)	20.796
Lower resistance limit (m ² -K/W)	20.796
U-Value surface to surface (W/m ² -K)	0.048
R-Value (m ² -K/W)	20.796
U-Value (W/m²-K)	0.048

CASE 2



ROOF SECTION DETAIL

Edit construction - PORSCHE ROOF

Constructions	
Layers	Surface properties
Image	Calculated
Cost	Condensation analysis
Inner surface	
Convective heat transfer coefficient (W/m ² -K)	4.460
Radiative heat transfer coefficient (W/m ² -K)	5.540
Surface resistance (m ² -K/W)	0.100
Outer surface	
Convective heat transfer coefficient (W/m ² -K)	19.870
Radiative heat transfer coefficient (W/m ² -K)	5.130
Surface resistance (m ² -K/W)	0.040
No Bridging	
U-Value surface to surface (W/m ² -K)	0.861
R-Value (m ² -K/W)	1.301
U-Value (W/m²-K)	0.769
With Bridging (BS EN ISO 6946)	
Thickness (m)	1.4920
Km - Internal heat capacity (KJ/m ² -K)	78.2400
Upper resistance limit (m ² -K/W)	1.301
Lower resistance limit (m ² -K/W)	1.301
U-Value surface to surface (W/m ² -K)	0.861
R-Value (m ² -K/W)	1.301
U-Value (W/m²-K)	0.769

CONSTRUCTION DEATILS LAYERS

CASE 2

Calculated Values	
Total solar transmission (SHGC)	0.117
Direct solar transmission	0.04536
Light transmission	0.102
U-value (ISO 10292/ EN 673) (W/m ² -K)	0.629
U-Value (W/m²-K)	0.665

CALCULATED VALUE SHGC

Lighting Template

Template LED

General Lighting

☒ On

Normalised power density (W/m²-100 lux) 2.5000

 Schedule PORSCHE SCHEDULE

Luminaire type 1-Suspended

Return air fraction 0.000

Radiant fraction 0.420

Visible fraction 0.180

Convective fraction 0.400

LIGHT TYPE DATA

HVAC Template

Template Packaged DX

Mechanical Ventilation

☒ On

Outside air definition method 4-Min fresh air (Sum per person + per area)

Operation

 Schedule PORSCHE SCHEDULE

Economiser (Free Cooling)

Heat Recovery

Auxiliary Energy

Heating

Cooling

☒ Cooled

 Cooling system Default

Fuel 1-Electricity from grid

Cooling system seasonal CoP 3.3

Supply Air Condition

Minimum supply air temperature (°C) 12.00

Minimum supply air humidity ratio (g/g) 0.0077

Cooling limit type 3-Limit flow rate and capacity

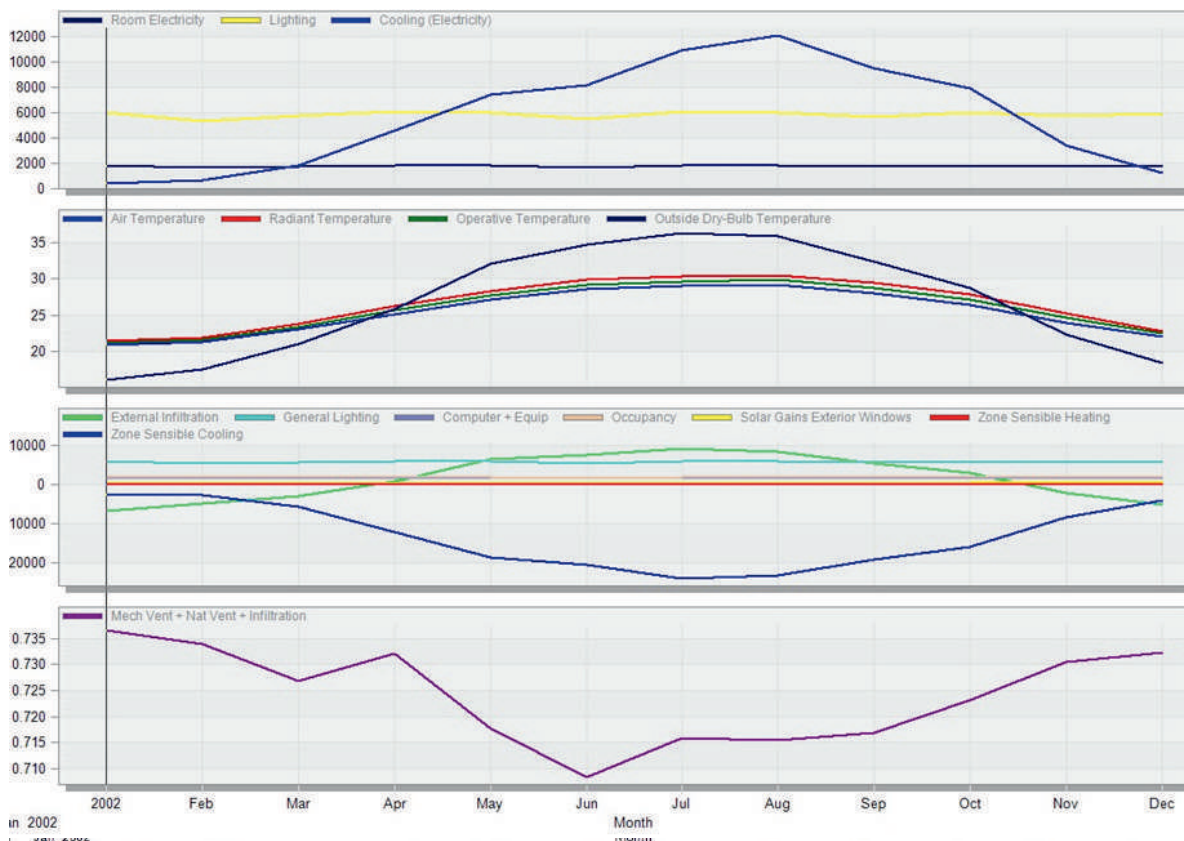
Operation

 Schedule PORSCHE SCHEDULE

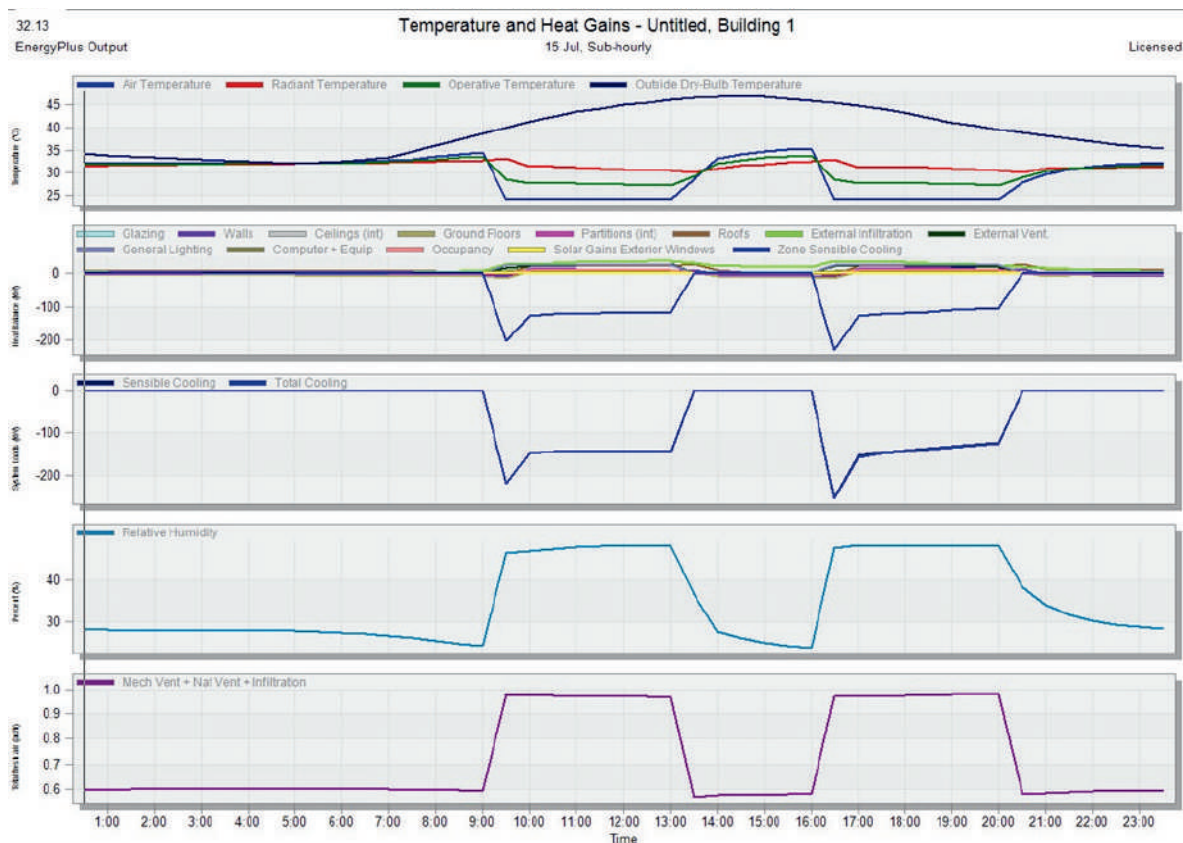
HVAC DATA



CASE 2



SIMULATION ANALYSIS



COOLING DESIGN ANALYSIS



CASE 2

Month	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Jan	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
Feb	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
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Apr	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
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Jun	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
Jul	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Copy of 9:00 - 13:00 & 16:00 - 20:00	Off	8:00-21:00	Copy of 9:00 - 13:00 & 16:00 - 20:00
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FIRST CASE REAL HVAC SCHEDULE

Month	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
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WORKING HOURS SCHEDULE

Site and Source Energy

	Total Energy [kWh]	Energy Per Total Building Area [kWh/m ²]	Energy Per Conditioned Building Area [kWh/m ²]
Total Site Energy	314885.23	367.53	367.53
Net Site Energy	314885.23	367.53	367.53
Total Source Energy	524020.55	611.63	611.63
Net Source Energy	524020.55	611.63	611.63

Utility Use Per Total Floor Area

	Electricity Intensity [kWh/m ²]	Natural Gas Intensity [kWh/m ²]	Additional Fuel Intensity [kWh/m ²]	District Cooling Intensity [kWh/m ²]	District Heating Intensity [kWh/m ²]	Water Intensity [m ³ /m ²]
Lighting	81.48	0.00	0.00	0.00	0.00	0.00
HVAC	0.00	0.00	0.00	261.61	0.00	0.00
Other	24.44	0.00	0.00	0.00	0.00	0.00
Total	105.92	0.00	0.00	261.61	0.00	0.00

End Uses

	Electricity [kWh]	Natural Gas [kWh]	Additional Fuel [kWh]	District Cooling [kWh]	District Heating [kWh]	Water [m ³]
Heating	0.00	0.00	0.00	0.00	0.00	0.00
Cooling	0.00	0.00	0.00	224133.71	0.00	0.00
Interior Lighting	99908.86	0.00	0.00	0.00	0.00	0.00
Exterior Lighting	0.00	0.00	0.00	0.00	0.00	0.00
Interior Equipment	20942.66	0.00	0.00	0.00	0.00	0.00
Exterior Equipment	0.00	0.00	0.00	0.00	0.00	0.00
Fans	0.00	0.00	0.00	0.00	0.00	0.00
Pumps	0.00	0.00	0.00	0.00	0.00	0.00
Heat Rejection	0.00	0.00	0.00	0.00	0.00	0.00
Humidification	0.00	0.00	0.00	0.00	0.00	0.00
Heat Recovery	0.00	0.00	0.00	0.00	0.00	0.00
Water Systems	0.00	0.00	0.00	0.00	0.00	0.00
Refrigeration	0.00	0.00	0.00	0.00	0.00	0.00
Generators	0.00	0.00	0.00	0.00	0.00	0.00
Total End Uses	90751.52	0.00	0.00	224133.71	0.00	0.00

Electric Loads Satisfied

	Electricity [kWh]	Percent Electricity [%]
Fuel-Fired Power Generation	0.000	0.00
High Temperature Geothermal*	0.000	0.00
Photovoltaic Power	0.000	0.00
Wind Power	0.000	0.00
Power Conversion	0.000	0.00
Net Decrease in On-Site Storage	0.000	0.00
Total On-Site Electric Sources	0.000	0.00
Electricity Coming From Utility	90751.520	100.00
Surplus Electricity Going To Utility	0.000	0.00
Net Electricity From Utility	90751.520	100.00
Total On-Site and Utility Electric Sources	90751.520	100.00
Total Electricity End Uses	90751.520	100.00

INTERIOR EQUIPMENT AND LIGHTING



ENERGY CONSERVATION

Cost schedule study

	ECN	Initial cost	Annual saving	Life span	NO. of years	
Doors	Change the door with automatic sliding door	11,250 SAR	4,203.21 SAR	15-20 Years	2.6 Years	37%
	AIR CURTAIN GARAGE	15,059.36 SAR	15,059.36 SAR	18 years	2.1 Year	46.7%
Windows	High efficiency triple glass low-e	134,078 SAR	6,304.65 SAR	20-50 Years	116 Years	6%
	High quality weather stripping	2,473 SAR	-----	20-50 Years	8.1 Years	
Walls	Cover the column with insulated material and gypsumboard	1,399.72 SAR	2101.44 SAR	15-20 Years	7.5 Month	150%
Roof	Paint the roof	66,375 SAR	5,617 SAR	15-20 Years	11 years	9%
Lighting	Upgrade the old LED lighting	15,040 SAR	2,081.442 SAR	17 Years	4.73Year	13.8%
A/C system	Simple shade structure as a metal or fabric canopy	8,000 SAR	6,304.32 SAR	10-12 Years	1.2 Years	78.8%
	Change some of the zones to concealed	412,000 SAR	8,712 SAR	15-20 Years	30 Years	78.8%
Equipment & findings	Use motion detector for tv	400 SAR	150.98 SAR	20 Year	2.64 Year	37.74

OPERATION & MAINTENANCE

Operation and maintenance

- Follow the schedule of the maintenance of the HVAC
- Turn off the devices after working hour and during the break time.
- Monthly maintenance instead of annual.
- Use the daylight effectively
- Do maintenance for the revolving door
- Do maintenance the burning lighting.

SUMMARY

- This section talks about analyzing how energy flows through a building. It mentions that the goal is to examine how energy enters and is used within the building, and then identify ways to improve energy efficiency.
- Main Goal: Optimize consumption of energy within the building.
- Secondary Goal: Achieve electricity cost savings.
- Objective: The report outlines a comprehensive approach to reducing energy consumption in a showroom, including a thorough inspection, analysis of historical data, and potential financing options.
- Purpose and Scope: The report assesses building energy consumption, identifies areas for improvement, investigates factors contributing to excessive use, proposes solutions, and models them using a building energy simulation program.



Operation and maintenance

Motion sensor for TV
HVAC maintenance
Lighting maintenance
Revolving Door maintenance
Equipment operation

ENERGY CONSERVATION

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Energy conservation measures

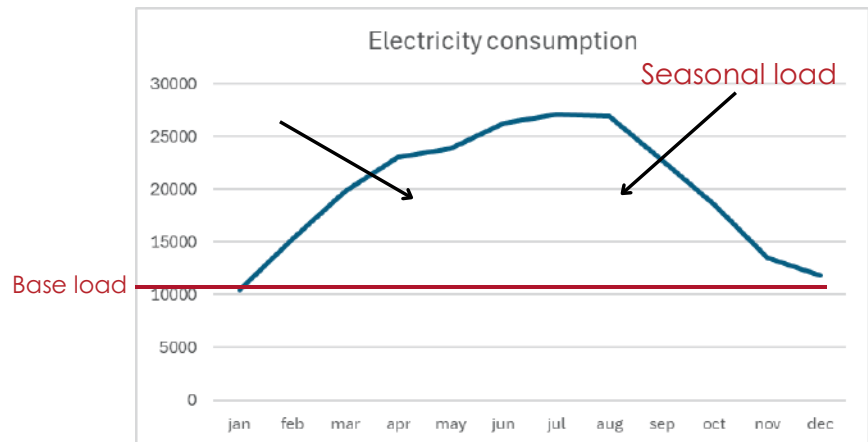
Add sliding door
Double garage
Add vertical shading devices
Change glazing type
Weatherstripping
Column cover



SUMMARY

Electricity Bill 2022

Month	KWh
Jan	10482
Feb	15278
Mar	19834
Apr	23012
May	23849
Jun	26214
Jul	27053
Aug	26910
Sep	22716
Oct	18516
Nov	13503
Dec	11826
Total	239193

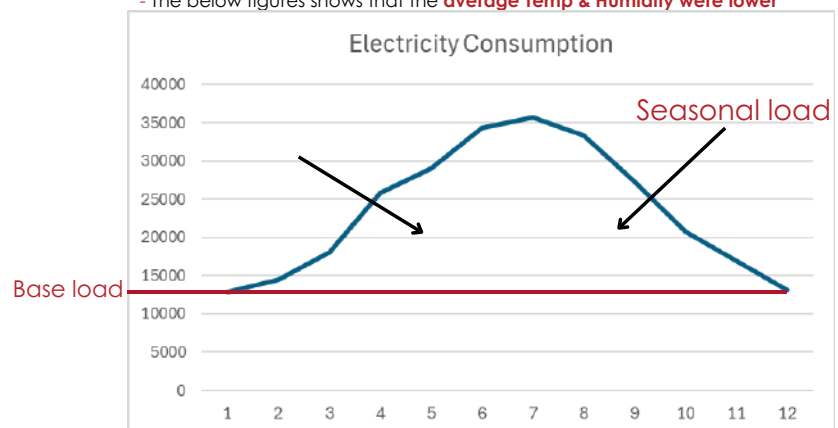


Causes:

- Based on Porsche analysis due to **global crisis** the car market dropped dramatically though that the inflation got higher.
- Employee number **dropped** so the electricity bill is directly proportional
- **Online purchasing** was booming reducing the visitors number to the showroom
- Due to **part shortage** the new car market dropped
- The below figures shows that the **average Temp & Humidity were lower**

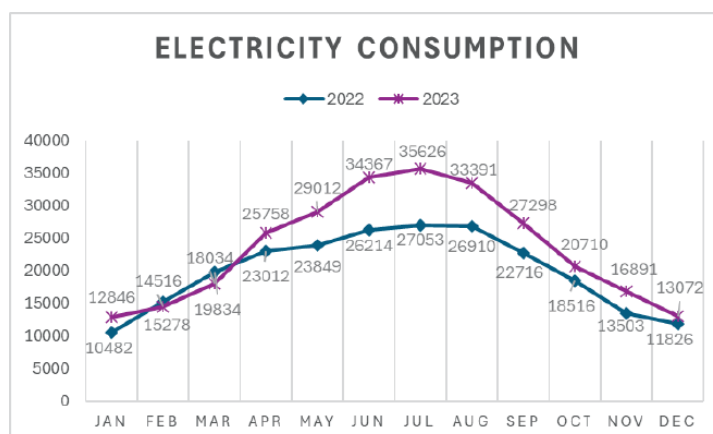
Electricity Bill 2023

Month	KWh
Jan	12,846
Feb	14,516
Mar	18,034
Apr	25,758
May	29,012
Jun	34,367
Jul	35,626
Aug	33,391
Sep	27,298
Oct	20,710
Nov	16,891
Dec	13,072
Total	281521



- The **Facility Manager Quoted**: At the end of 2023 we found an issues with the HVAC system, The first issue was leakage in the ducts. **Secondly** two of the Package units weren't up to their performance.

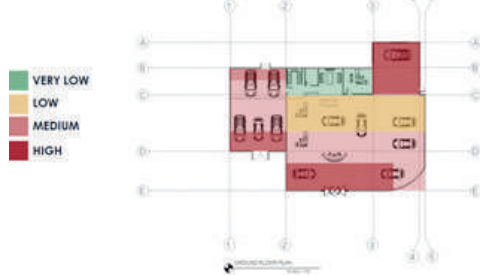
- The number of **employment increased**
- Events took place in the showroom 2023 (**The reveal of the new Taycan**).



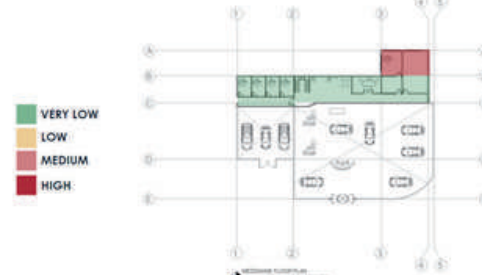
The graph shows the difference between 2022 & 2023

SUMMARY

AREA OF HIGH ENERGY USED



GROUND FLOOR



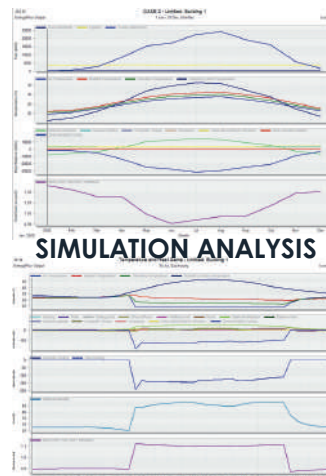
MEZZANINE FLOOR

RECOMMENDATION

Window	- Change glazing type (low-e coating, high efficiency Triple glass, coloured glass). - higher quality weather stripping.
Walls	- Increase the insulation - Close the air gaps - Weather the cladding infiltration fix the wall cracks.
Roof	- Paint the roof with lighter colours. - Use PV-system to use it during the break time to save electricity. - Double roof system.
Lighting	- Turn off the light during the break time. - Use a type of lighting that suitable for the show room. - Increase the number of lighting in the show room - Upgrade the old led light (20w) with a (40w-60w). - Putting the focal point under the sky light to use the natural lighting.
A/C system	- We can put shade above them to help with the performance and help maintain the system. - Change the system to concealed. - Change the ductwork and the leakage problems
equipment loads findings	- Installing a motion detector so that when no one is around, it will automatically turn off and on when someone walks by. - all electric devices must be unplugged during off hours.

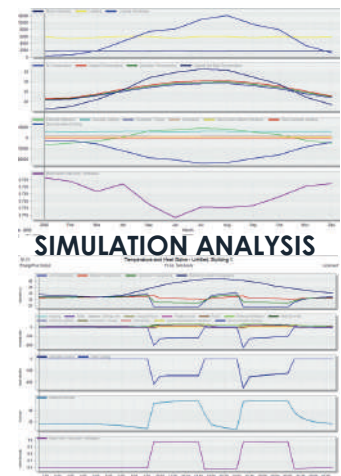
CASES SIMULATION

CASE 1



COOLING DESIGN ANALYSIS

CASE 2



COOLING DESIGN ANALYSIS

CONCLUSION

This **energy audit** has identified a number of opportunities for the Porsche showroom in Khobar, Saudi Arabia to improve its energy efficiency and reduce its operating costs. By implementing the recommendations outlined in this report, the showroom can achieve significant **cost savings** while also contributing to a more sustainable future.

We are confident that by following the recommendations in this report, the Porsche showroom can achieve its energy-saving goals while maintaining a comfortable and luxurious environment for its customers and staff. We are also available to provide further assistance and guidance throughout the implementation process



DONE BY:

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Saeed Abu Almakarem [REDACTED]

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